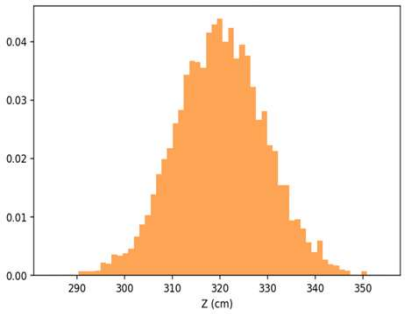
Agenda

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Last time: Why normal?
  - z-score and proportions
  - Sums of
    - independent but
    - not identical
    - normal distribution
  - The sum is normal
  - Central Limit Theorem



2

Sums of independent normal distribution

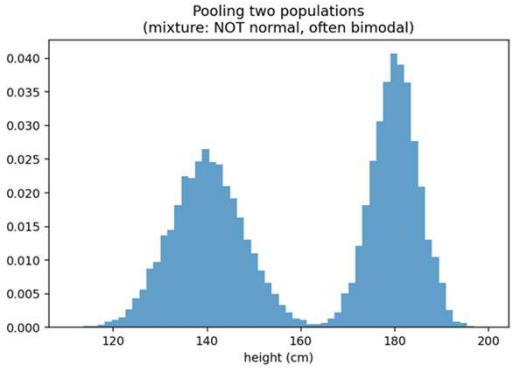
- $Y = X_1 + X_2 + \cdots X_n$ where each $X_i \sim N(\mu_i, \sigma_i^2)$ and *independent*
  - Answer: $Y \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$
- Per theorem,
  - What happens if measure heights X_1 of 100 selected national basketball players (normal distribution)
  - and, say, 100 kho-kho players X_2 (also normal distribution)
  - Assume independence
  - Put them all in this room
  - Distribution?

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

3

Sums of independent normal distribution

- $Y = X_1 + X_2 + \cdots X_n$ where each $X_i \sim N(\mu_i, \sigma_i^2)$ and independent
- Answer: $Y \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$
- Per theorem,
  - what happens if measure heights X_1 of 100 selected national basketball players
  - and, say, 100 kho-kho players X_2?
  - Put them all in this room
  - Distribution?



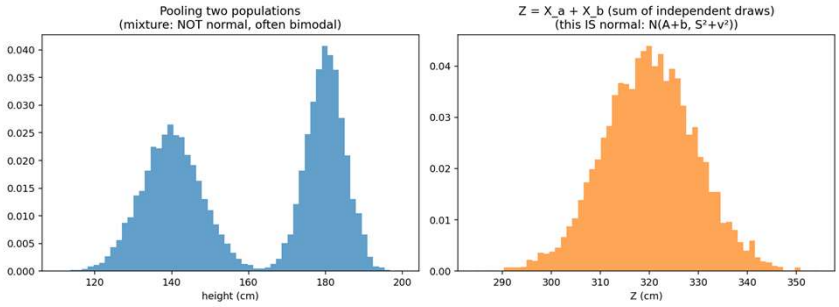
5

Sums of independent normal distribution

- $Y = X_1 + X_2 + \cdots X_n$ where each $X_i \sim N(\mu_i, \sigma_i^2)$ and independent
- Answer: $Y \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$
- Per theorem, what happens if measure heights X_1 of national basketball players and, say, kho-kho players X_2?

The result is not about concatenation, or mixture

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

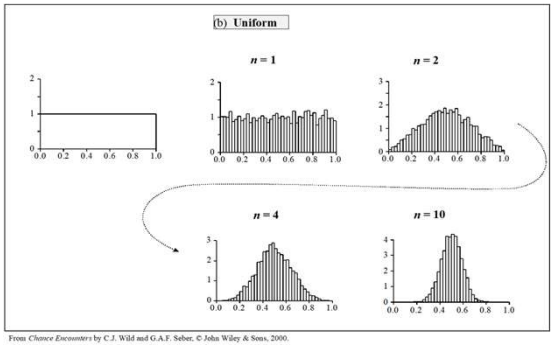


6

Agenda

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Last time: Why normal?
  - z-score, proportions
  - Sums of independent but not identical normal distribution
  - Sums of independent identical distributions, not necessarily normal (Central Limit Theorem)



From *Chance Encounters* by C.J. Wild and G.A.F. Seber, © John Wiley & Sons, 2000.

7

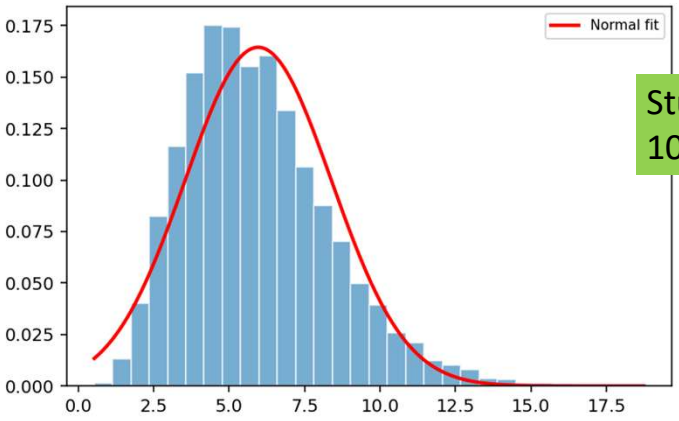
Why normal?

- For independent identical random variables, and $Y = X_1 + X_2 + \cdots X_n$
- Central limit theorem
  - For large n, the distribution of Y(n) is approximately normal with mean and variance as earlier seen as if the distributions were normal
- How large is n?

The statement of the theorem is not precise

8

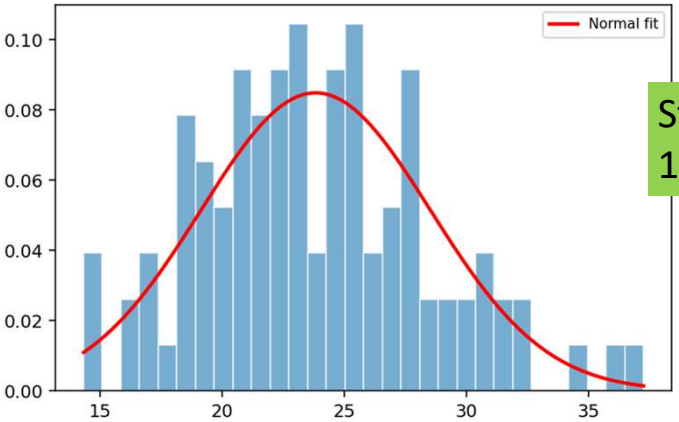
Which distribution is “more” normal?



Student A: n = 6, 10000 experiments

9

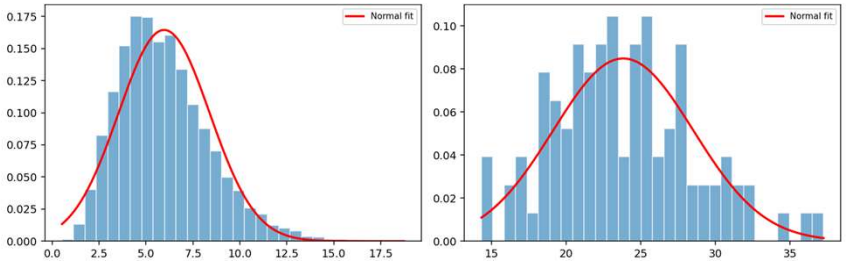
Which distribution is “more” normal?



Student B n=24, 100 experiments

10

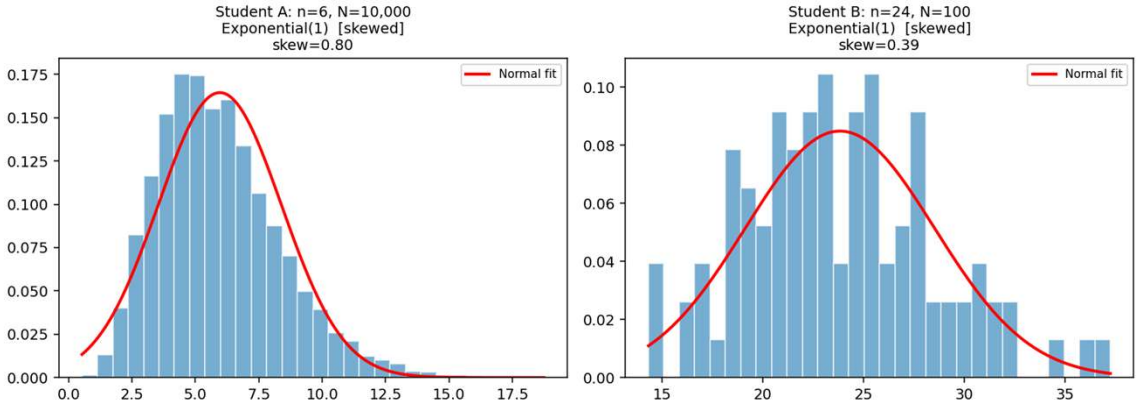
Which distribution is “more” normal?



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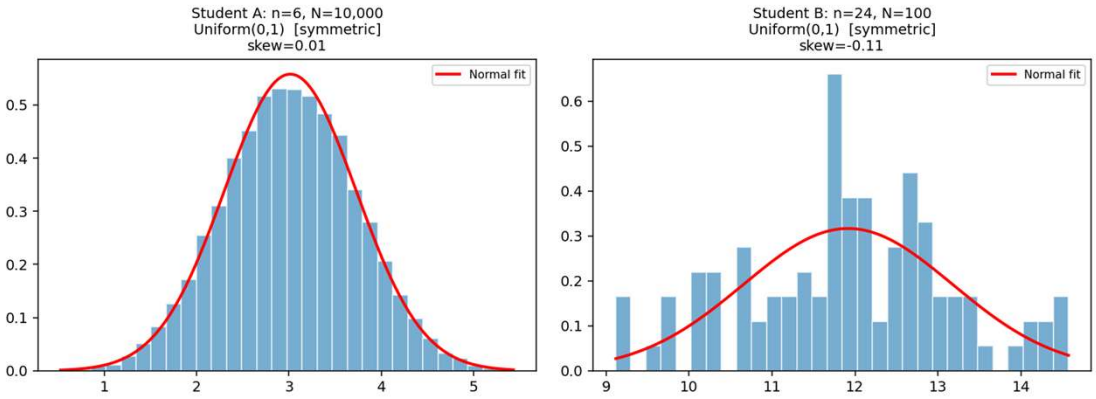
11

Which distribution is “more” normal?



13

But …



14

Why normal?

- For independent identical random variables, and $Y = X_1 + X_2 + \cdots X_n$
- Central limit theorem
  - For large (but finite) n, the distribution of Y is approximately normal with mean and variance as earlier seen as if the distributions were normal
  - Finite n, sample mean $\overline{x} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$
  - Notice that it helps to know the mean and variance of typical distributions corresponding to X_i

Takeaway message

15

Normal and Sums of Bernoulli

- Let X= $X_1 + X_2 + \cdots X_n$
  - Each $X_i \sim \text{Bern}(p)$
- Clearly $X \sim \text{Bin}(n, p)$
- Exact distribution
- Central limit theorem approximation:
  - For large n, $X \sim \mathcal{N}(np, np(1-p))$
  - $\frac{Y-np}{\sqrt{np(1-p)}}$ is standard normal

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

$$E(X) = np$$

$$\text{Var}(X) = np(1-p)$$

16

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

Example: Normal and Binomial



- Your CR is organizing “garba” in new-new-sac (100 size)
  - From past experience, students who register turn up with a probability of 0.7
  - Your CR wants to “overbook” the event maximizing cover fee
  - How many registrations can he accept if the probability of more than 100 showing up is 5%?
- X number of students: $\text{Bin}(n, 0.7)$
  - What is the largest n for “good case” $\longrightarrow P(X \geq 101) \leq 0.05$)

17

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

Example: Normal and Binomial

- Your CR is organizing "garba" in new-new-sac (100 size)
  - From past experience, students who register turn up with a probability of 0.7
  - Your CR wants to "overbook" the event maximizing cover fee
- X number of students: Bin$(n, 0.7)$
- What is the largest n for "good case" $\longrightarrow$ $P(X \geq 101) \leq 0.05)$
- Binomial: For each n,
  - Evaluate $P(X \geq 101) = \sum_{k=101}^{n} \binom{n}{k} (0.7)^k (0.3)^{n-k}$
  - For example, with n = 131, $P(X \geq 101) = \sum_{k=101}^{131} \binom{131}{k} (0.7)^k (0.3)^{131-k} \approx 0.0441$

18

Example: Normal and Binomial

$$E(X) = np$$

$$\mathrm{Var}(X) = np(1-p)$$

- How many registrations can he accept if the probability of more than 100 showing up is 5%?
  - $P(X \geq 101) \leq 0.05)$
- Normal approximation $X \approx \mathcal{N}(0.7n, 0.21n)$
  - Continuity correction
  - Map to z-score
  - Inverse probability and one-sided tail
  - Solve quadratic equation
  - Complicated binomial sum has same answer as quadratic equation

$$P(X \geq 101) \approx P(Y > 100.5),$$

$$P\left(Z > \frac{100.5 - 0.7n}{\sqrt{0.21n}}\right) \leq 0.05.$$

$$P(Z > 1.645) = 0.05,$$

$$\frac{100.5 - 0.7n}{\sqrt{0.21n}} \geq 1.645.$$

$$n \approx 131.24.$$

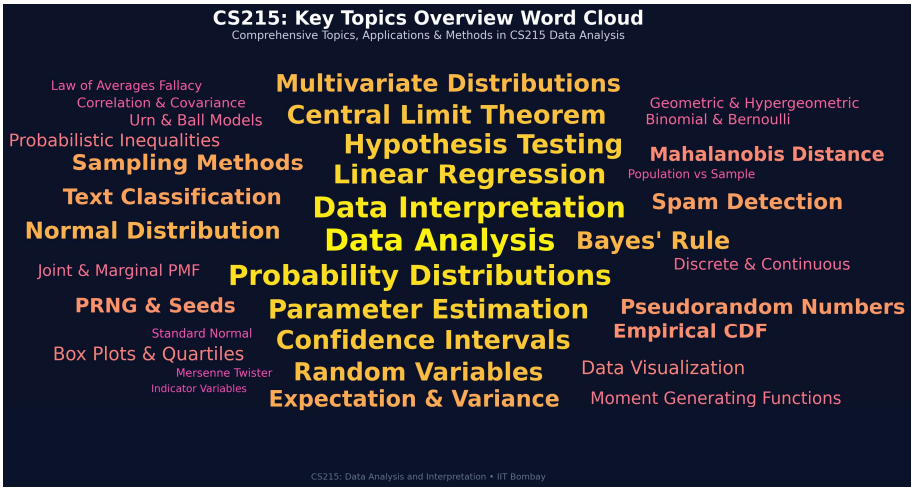
19

What topics have we discussed?

- Up to chapter 6 in the textbook (Sheldon Ross)
- What is the syllabus for the exam?
  - Coming up

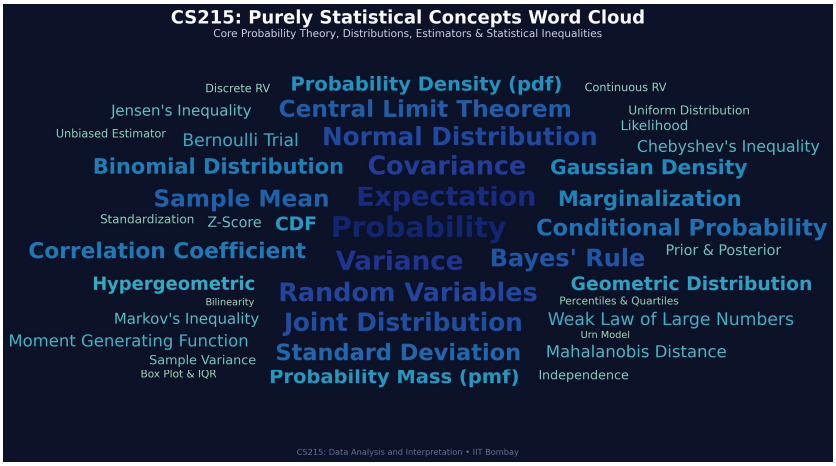
2

We have come a long way



3

We have come a long way



4

Let’s revisit the first week of classes

5

FOUNDATIONS

Population

POPULATION

The full process or distribution

Parameters: μ and σ^2
Fixed—but usually unknown

- Suppose someone (an "oracle") told you that values are
  - Only (W= 0 or W= 2 or W= 4)
  - What's the mean and variance?
  - Expected value: $\mu = 2$
  - Variance as $\sigma^2 = 8/3$

Basic "first order" knowledge you must have

1

6

FOUNDATIONS

Population and Observations

POPULATION

The full process or distribution

Parameters: μ and σ^2
Fixed—but usually unknown

observe

SAMPLE

Observed values: $x_1, \ldots, x_n$

Statistics: $\bar{x}$ and S^2
Change from sample to sample

Preparing for later courses: Inference on the population from the data

Basic "first order" knowledge you must have

1

7

Basic "first order" knowledge you must have

Example

- The weights of men has a mean of 70 kg with a variance of 2 kg
- Want mean and variance of the *total of 3 randomly* sampled men?
  - Assume independence, weights are normally distributed, let X be the weight of a randomly sampled man and T the total weight
  - A student wrote: T = 3X
  - Using the properties discussed in class, $E[T] =?$ $\text{Var}[T] = ?$

$$X \sim N(\mu, \sigma^2) \text{ and } Y = aX, \implies Y \sim N(a\mu, a^2\sigma^2)$$

8

FOUNDATIONS

Population and Observations

POPULATION

The full process or distribution

Parameters: μ and σ^2
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observe

SAMPLE

Observed values: $x_1, \ldots, x_n$

Statistics: $\bar{x}$ and S^2
Change from sample to sample

$$\overline{x} = \frac{\sum x_i}{n} \qquad \frac{\sum (x_i - \overline{x})^2}{n-1}$$

Preparing for later courses: Inference on the population from the data

Basic "first order" knowledge you must have

1

11

Experiments

Focusing on the first column, how do we use the language of mathematics to express the first column

All possible samples of size 2	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum(x-\bar{b})^2}{n-1}$	$\frac{\sum(x-\bar{b})^2}{n}$
0 0	(0+0)/2 = 0	$((0-0)^2+(0-0)^2)/1$ = 0	0
0 2	1	$((0-1)^2+(2-1)^2)/1$ = 2	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

- Does the first column represent
  - A: Two samples from one random variable W?
  - B: One sample each from two instances of the random variable W?

12

Experiments

Focusing on the first column, how do we use the language of mathematics to express the first column



All possible samples of size 2	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum(x-\bar{b})^2}{n-1}$	$\frac{\sum(x-\bar{b})^2}{n}$
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0 2	1	$((0-1)^2+(2-1)^2)/1$ = 2	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
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FOUNDATIONS

Population and Observations

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Observed values: $x_1, \ldots, x_n$

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Preparing for later courses: Inference on the population from the data

Basic "first order" knowledge you must have

1

14

Experiments

Focusing on the first column

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0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

- Assume the sample was of size 1
- What is the statistician's view of the expected value of the random variable W?

15

The weak law of large numbers

- Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed random variables, each with mean μ. For any $\varepsilon > 0$:

$$P\left\{\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right\} \to 0 \text{ as } n \to \infty$$

- The **sample mean** converges in probability to the true mean

16

16

Experiment

Now focusing on the second column

All possible samples of size 2	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum(x-\bar{b})^2}{n-1}$	$\frac{\sum(x-\bar{b})^2}{n}$
0 0	(0+0)/2 = 0	$((0-0)^2+(0-0)^2)/1 = 0$	0
0 2	1	$((0-1)^2+(2-1)^2)/1 = 2$	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

- Considering samples of size 2:
  - What will be statistician view of the expected value of the random variable?

17

Experiment

Now focusing on the last two columns

All possible samples of size 2	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum(x-\bar{b})^2}{n-1}$	$\frac{\sum(x-\bar{b})^2}{n}$
0 0	(0+0)/2 = 0	$((0-0)^2+(0-0)^2)/1 = 0$	0
0 2	1	$((0-1)^2+(2-1)^2)/1 = 2$	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

- Considering samples of size two,
  - Dividing by n results in $E[\overline{s}^2] = \frac{12}{9} = \frac{4}{3}$
  - Division by n underestimates the true variance
  - Dividing by (n-1) results in $E[\overline{s}^2] = \frac{24}{9} = \frac{8}{3}$

18

CLT

- As stated, the random variables X_i are independent and identically distributed.
- Alternate: require independence but allow the random variables to belong to different distributions
  - Technical conditions (e.g. variance of no one random variable, or groups of random variables is allowed to dominate the sum of the variances)
  - "take home message" … it works

○ Misconceptions

- "Lots" of random sampling "reveals" normal distribution
- Does not apply to random sampling of any variable
  - Applies to sampling distribution of mean
- Empirical rule of thumb (n=30) can lead to seriously flawed inferences

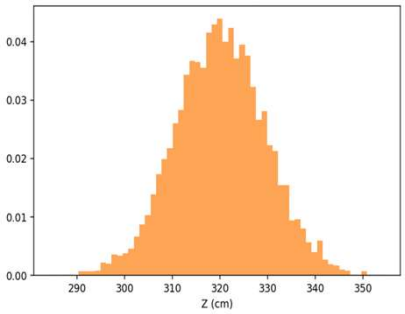
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24

Agenda

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Last time:
  - z-score and proportions
  - Sums of independent but not identical normal distribution
  - Central Limit Theorem
    - "Common man" version
    - Technical version

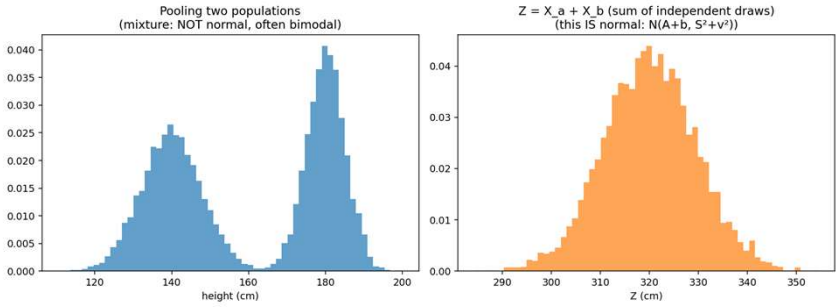


2

Sums of independent normal distribution

- $Y = X_1 + X_2 + \cdots X_n$ where each $X_i \sim N(\mu_i, \sigma_i^2)$ and independent
  - Answer: $Y \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$
- Per theorem, what happens if measure heights X_1 of national basketball players and, say, kho-kho players X_2?

The result is not about concatenation, or mixture



3

Why normal?

- For independent identical (iid) random variables, and $Y = X_1 + X_2 + \cdots X_n$
- Central limit theorem
  - For large (but finite) n, the distribution of Y is approximately normal with mean and variance as earlier seen as if the distributions were normal
  - Finite n, sample mean $\overline{x} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$
    - Take home message emphasizes the division by n
    - Although Y above is actually Y_n

Take home message

4

Normal and Sums of Bernoulli

- Let X= $X_1 + X_2 + \cdots X_n$
  - Each $X_i \overset{\text{iid}}{\sim} \text{Bern}(p)$
- Clearly $X \sim \text{Bin}(n, p)$
  - We know exact distribution
- Central limit theorem approximation:
  - For large n, $X \sim \mathcal{N}(np, np(1-p))$
  - $\frac{\text{X}-np}{\sqrt{np(1-p)}}$ is standard normal

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

5

Central limit theorem

More precise statement

- $X_1, X_2, \ldots, X_n$ independent and identically distributed random variables each with mean μ and variance $\sigma^2 < \infty$
  - $Y_n \triangleq X_1 + X_2 + \cdots X_n$
- As $n \to \infty$, the distribution Y_n converges to that of $\mathcal{N}(0, \sigma^2)$
  - Y_n converges in distribution to normal

The statement is more precise but a tad awkward

Convergence in distribution (see here for more)

$$Y_n = \sqrt{n}\left(\frac{\sum_{i=1}^{n} X_i}{n} - \mu\right)$$

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6

CLT

Why the complicated statement?

$$Y_n = \sqrt{n}\left(\frac{\sum_{i=1}^{n} X_i}{n} - \mu\right) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

Untangling

$$Y_n = \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right) \dot{\sim} N(0, \sigma^2)$$

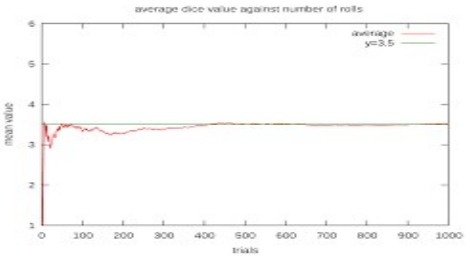
$$\frac{1}{n}\sum_{i=1}^{n} X_i - \mu \;\dot{\sim}\; N(0, \sigma^2/n) \qquad X \sim N(\mu, \sigma^2) \text{ and } Y = aX, \implies Y \sim N(a\mu, a^2\sigma^2)$$

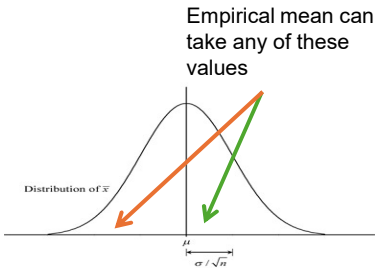
$$\frac{1}{n}\sum_{i=1}^{n} X_i \;\dot{\sim}\; N(\mu, \sigma^2/n)$$

7

CLT and LLN

- The law of large numbers says that the empirical mean calculated from a large number of samples is equal to (or very close) to the true mean μ (of the distribution from which the samples were drawn)
- The central limit theorem says that the empirical mean calculated from a large number of samples is a random variable following a Gaussian distribution with mean equal to the true mean μ



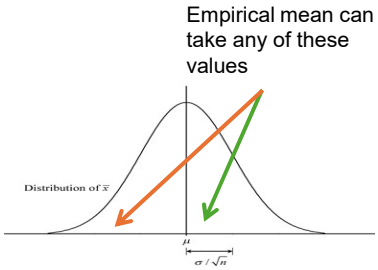


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8

CLT and LLN

- Limiting argument as $n \to \infty$
  - The variance drops to zero



$$Y_n = \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right) \dot{\sim} N(0, \sigma^2)$$

$$\frac{1}{n}\sum_{i=1}^{n} X_i - \mu \;\dot{\sim}\; N(0, \sigma^2/n) \qquad X \sim N(\mu, \sigma^2) \text{ and } Y = aX, \implies Y \sim N(a\mu, a^2\sigma^2)$$

$$\frac{1}{n}\sum_{i=1}^{n} X_i \;\dot{\sim}\; N(\mu, \sigma^2/n)$$

83

9

The weak law of large numbers

- Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed random variables, each with mean μ. For any $\varepsilon > 0$:

$$P\left\{\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right\} \to 0 \text{ as } n \to \infty$$

- The **sample mean** converges in probability to the true mean

Must know

10

10

CLT

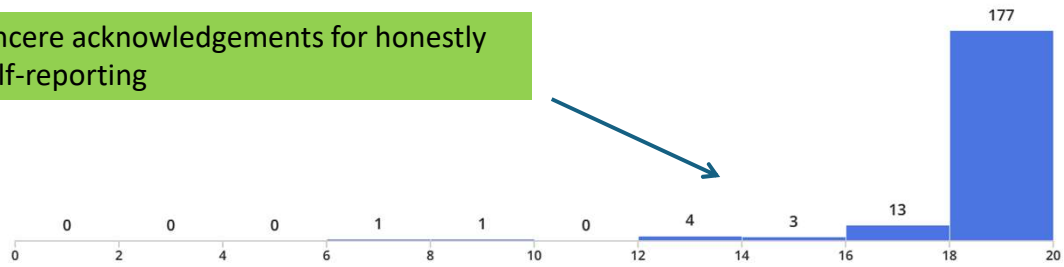
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  - Technical conditions (e.g. variance of no one random variable, or groups of random variables is allowed to dominate the sum of the variances)
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79

11

Part A results

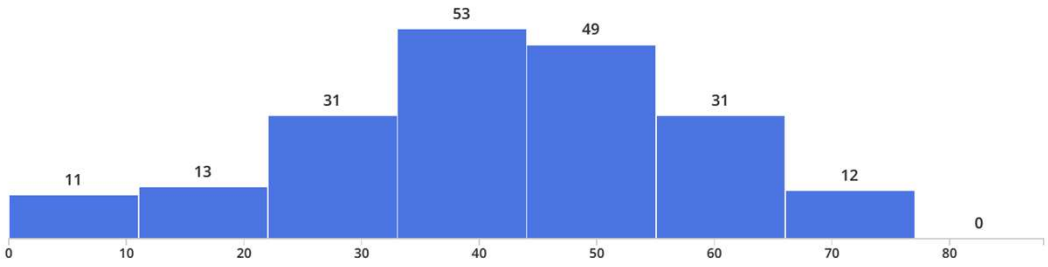
Sincere acknowledgements for honestly self-reporting



Minimum	Median	Maximum	Mean	Std Dev
6.0	20.0	20.0	19.01	1.92

2

Part B



Minimum	Median	Maximum	Mean	Std Dev
0.0	42.0	73.0	40.95	16.36

3

Part B Instructions

- You may assume any result in the Piazza-provided lecture handouts as “known”, i.e., not to be proved.
- Conversely, you cannot assume results
  - that have not been presented in class or posted in the handouts.
  - that are implicit or explicit in homework solutions
  - Such results are to be derived if needed during the exam.
- All questions require precise and complete mathematical arguments to earn full credit

4

Agenda

- Midsemester Exam
  - Self evaluation
  - Menti
  - Question
  - Answers
    - Piazza: common mistakes, imprecise answers, incorrect answer, partial marking
  - Results
  - Display of answer papers
    - Regrade request process
    - Friday, Students with Low Marks,

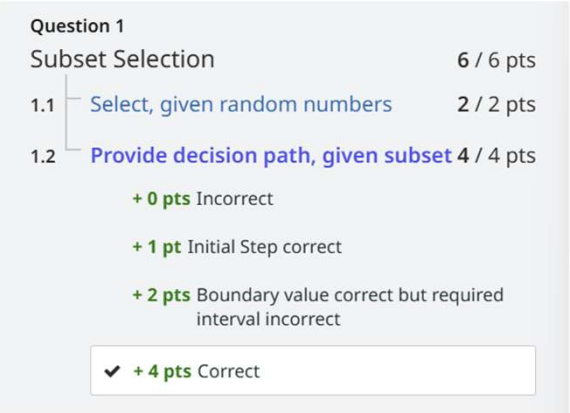
5

Subset Selection

- {B,D,A,E,C,F,G,H}
  - 0.52, 0.57, 0.66, 0.30, 0.80, 0.90, 0.25, 0.70
- Correct subset {D,E,G,H}
  - 4/8, 4/7 ~= 0.571, 3/6, 3/5, 2/4, 2/3
- Decision path for subset {D, E, F, H}
  - [0.50, 1): B Not selected, (0, 4/7): D: Selected, [0.5, 1) A not selected
  - (0, 3/5): E: Selected, [0.5, 1): C: Not Selected, (0, 0.66) F: Selected
  - [0.50, 1): G Not selected, H: Selected (Forced)

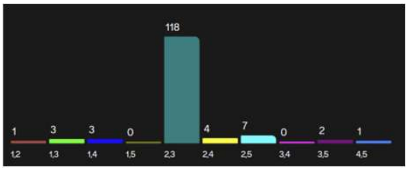
6

Grading Interface



7

Results



1: Subset Selection 6.0 points

1.1: Select, given random numbers 2.0 points 66%

Click to Add Tags

1.2: Provide decision path, given subset **4.0 points** 63%

Click to Add Tags

8

Attendance requirement

- 44 scheduled classes, 80% threshold met implies DX grade cannot be given for lack of attendance
- Identify the binomial parameters
  - Binomial(21, 0.75)

Student	Classes attended
A	16
B	18
C	20

9

Attendance requirement

Student	Classes attended
A	16
B	18
C	20

- 44 scheduled classes, 80% threshold met implies DX grade cannot be given for lack of attendance
- Identify the binomial parameters
  - Binomial(21, 0.75)
- Student A will not meet threshold if he attends 0 classes, or 1 classes, or 2 classes, and so on up to 17 classes since 36-16-2=18
- P(X <= 17) = 0.808 which is greater than the 4% threshold specified

10

Results



11

Red and Blue Balls in Urns

- One of two urns, A or B, is chosen with equal probability. Urn A contains 3 red balls and 1 blue ball, while urn B contains 1 red ball and 3 blue balls. Balls are drawn independently **with replacement** from the chosen urn.
- If the first two balls drawn are red, find the probability that urn A was chosen.
- Suppose that among the first n draws, exactly r are red. Find the condition on r and n under which urn A is more likely than urn B.

12

Results



13

Jenny and IPL franchise

- X and Y are independent normal with the same mean (70) and variance (64)
  - Want to understand S = X+Y and D=X-Y
- Find $\mathbb{E}[e^{uS+vD}]$ and simplify (*state assumptions you make*)
- Find $\mathrm{Cov}(S, D)$
- Find $\Pr(D \geq 8 | S > 150)$. Unjustified assumptions receive no credit

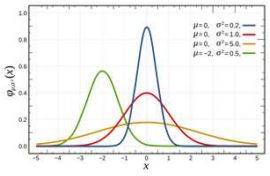
14

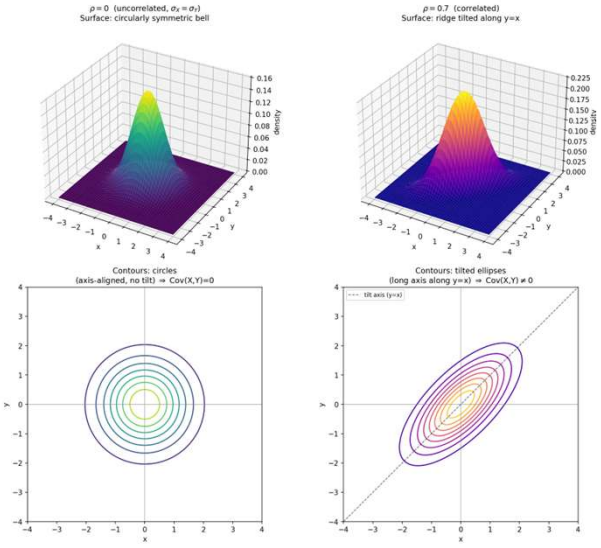
Results

4: IPL	13.0 points		
4.1: Expected Value of e^(uS + vD)	**6.0 points**		65%
Click to Add Tags			
4.2: Cov(S,D)	3.0 points		79%
Click to Add Tags			
4.3: Probability of difference > 8 given sum > 150	4.0 points		0%
Click to Add Tags			

15

Joint distributions of S and D





In both cases, any slice of the "Mexican" hat will give us a nice Gaussian

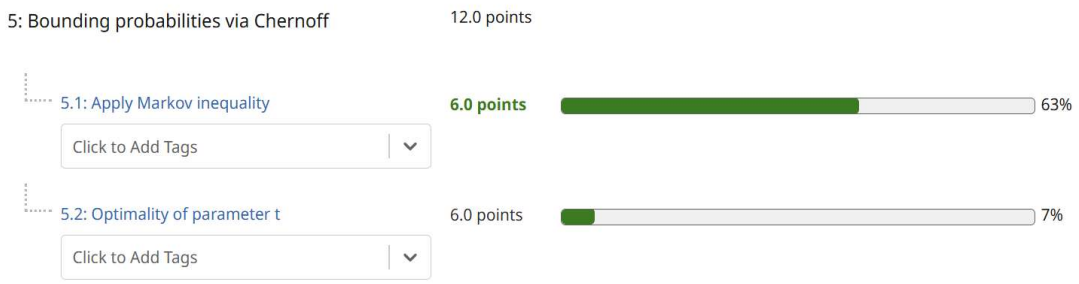
16

Bounding probability of the binomial

- We know the pmf of the binomial, but what about the cdf?
- Simpler question: What is the probability that in *n* trials we get more than *np* successes (e.g. *nq*, with $0 < p < q < 1$)
  - Standard Markov $\Pr(S_n \geq nq) \leq \frac{\mathbb{E}[S_n]}{nq}$
  - With p = 0.2, q=0.5, direct Markov gives 0.4 (no dependance on n)
  - With $np < a < n$ and $q = \frac{a}{n}, p < q < 1$. Show
    - $P(S_n \geq nq) \leq \left[\left(\frac{p}{q}\right)^q \left(\frac{1-p}{1-q}\right)^{1-q}\right]^n$
- This is the tightest Chernoff bound (in the example 0.8^n)

17

Results



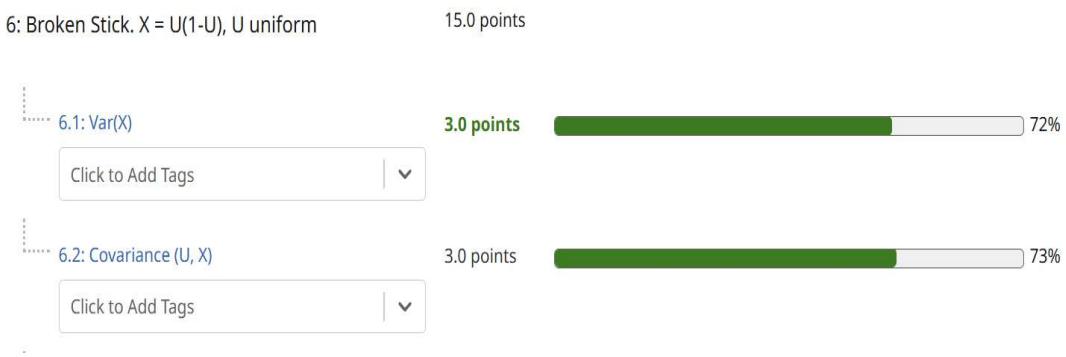
18

Stick

- Stick broken into two: X = U(1-U)
  - Expectation, Variance, Covariance (X, U)

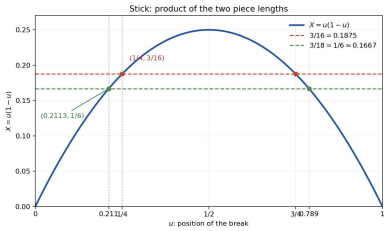
19

Results



20

Stick



- Stick broken into two: X = U(1-U)
  - Earlier, covariance = 0 gave us suspicion of independence, now covariance = 0 asks us to check independent or not?
  - Umm X = f(U) that sounds suspiciously like not independent
    - Want $\Pr(X = a|U = u) \stackrel{?}{=} \Pr(X = a)$?
  - We want to look for events that knowing something about X makes U to be constrained, it can't be whatever it wants
    - Argue about the event $\Pr(X \geq 3/16|U < 1/4)$

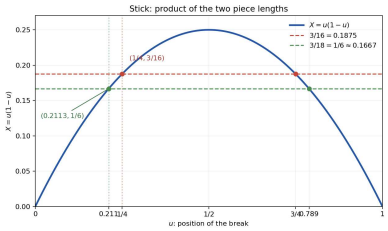
21

Results



22

Stick



- Stick broken into two: X = U(1-U)
  - Earlier, covariance = 0 gave us suspicion of independence, now covariance = 0 asks us to check independent or not?
  - Wait, there is more! show that X and Y are independent
    - Y=1 if U < 0.5
    - This appears to be unexpected!

23

Results

6.4: Independence of X and Y **(only 99% graded)** 6.0 points 10%

Click to Add Tags

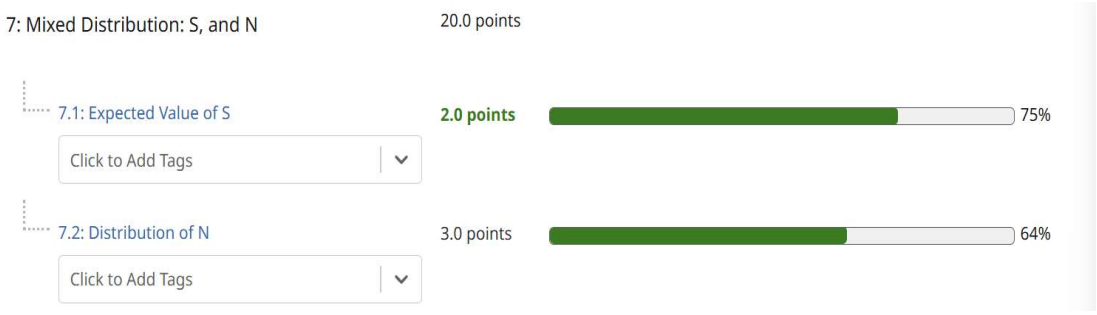
24

Mixed Distribution

- S = sum of i.i.d. X_i(continuous, symmetric)
- N counts the number of such $X_i$which is positive
- Find
  - $\mathbb{E}[S]$
  - Distribution of N

25

Results



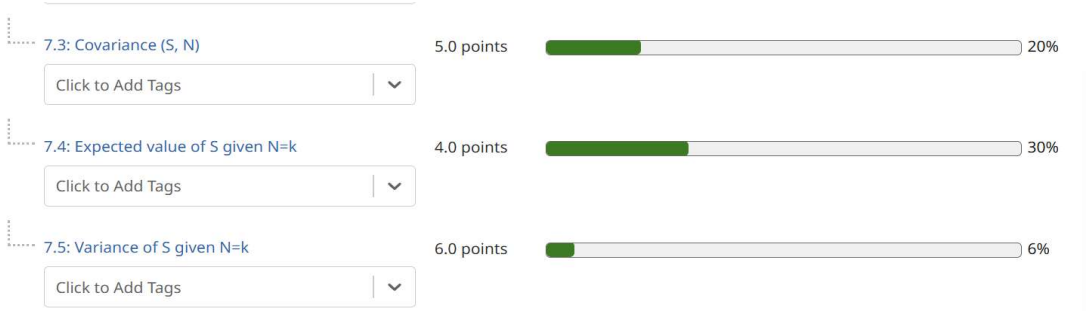
26

Mixed Distribution

- S = sum of i.i.d. X_i(continuous, symmetric)
- N counts the number of such $X_i$which is positive
- Find
  - $\mathrm{Cov}(S, N)$
  - $\mathbb{E}[S \mid \mathrm{N} = \mathrm{k}]$
  - $\mathrm{Var}(S|N = k)$

27

Results



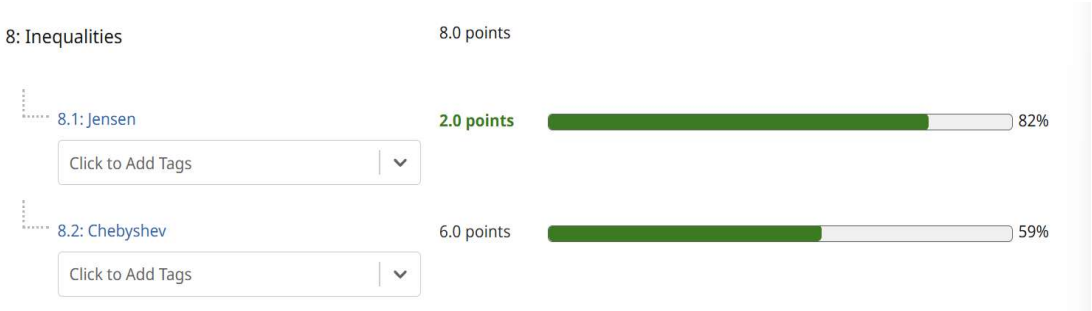
28

Chebychev

- X is random with expectation μ and finite variance
- Jensen's inequality
  - For a convex function, $f\big(\mathbb{E}(x)\big) \leq \mathbb{E}[f(x)]$
- For $0 < \theta < 1$, $\Pr(X \leq \theta\mu) \leq$?

29

Results



30

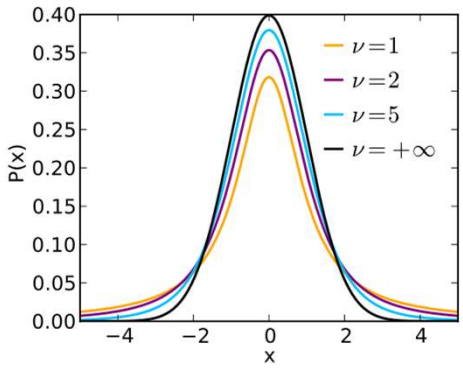
Agenda

- Midsemester Exam
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    - Piazza: common mistakes, imprecise answers, incorrect answer, partial marking
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  - Display of answer papers
    - Regrade request process
    - Friday, Students with Low Marks,

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Agenda

- Last time
  - Central Limit Theorem
- Today
  - Estimates and Estimators
  - Means and proportions in the context of CLT
  - Standard Error and Approximations
  - The "student" distribution



1

Expectation and Variance of iid

- For identical X_i random variables, with $\mathbb{E}[X_i] = \mu$
  - $Y = X_1 + X_2 + \cdots X_n$ and $\bar{X} = Y/n$, $\mathbb{E}[\bar{X}] = \mu$
  - With independence, and finite variance σ^2, $\text{Var}(\bar{X}) = \sigma^2/n$
- Proportions
  - In the 1998 Teen survey, 1000 American teenagers (12 – 17 years) were interviewed. 27% reported having seen illegal drugs sold at school
  - Although this sort of thing is a hypergeometric distribution, we model this is W~ Bin(n, p) (why?)
    - Here W = Bin (1000, 0.27)
    - The random variable $\hat{P}$ = W/n represents proportions
  - $\mathbb{E}[\hat{P}] = \text{p}, \text{Var}(\hat{P}) = \text{p}(1-\text{p})/n$

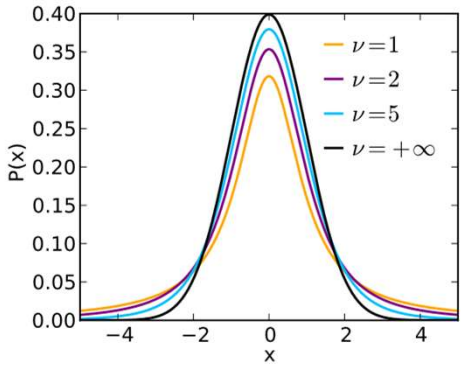
2

Parameter and estimates

- A ***parameter*** is a numerical characteristic of a population or distribution. It often is used to index the distribution
- You don't observe the whole population, you get a **random sample.**
  - A **statistic** is a random variable and a function of the sample random variables $X_1,...,X_n$ that does not depend on unknown parameters *before* observation
- **Estimator**: A statistic $T(X_1,...,X_n)$ that is specifically being used as a guess for an unknown parameter θ
- After you actually observe, an ***estimate*** is a realized quantity calculated from the data to estimate an unknown parameter
  - Capital letters refer to random variables
  - Small letters refer to observed values

3

CLT Approximation



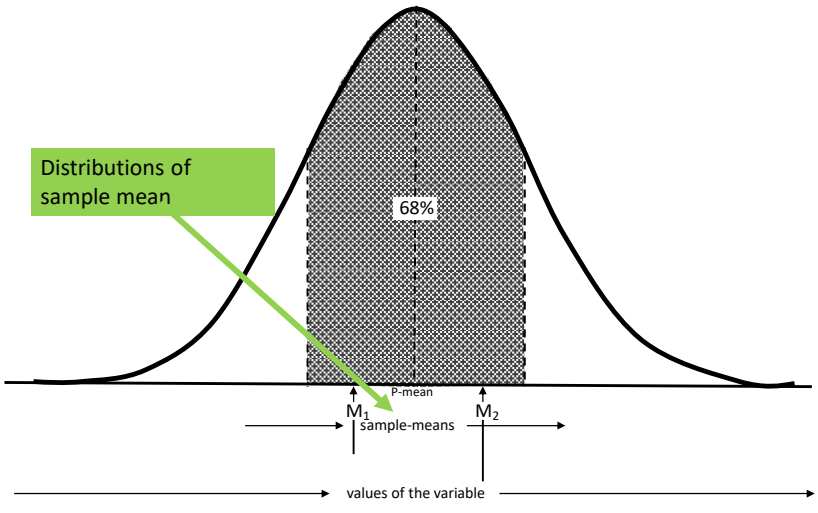
4

CLT

- For independent identical (iid) random variables, and $Y = X_1 + X_2 + \cdots X_n$ with $\mathbb{E}[X_i] = \mu$ and finite variance σ^2
- Central limit theorem
  - For large (but finite) n, the distribution of Y is *approximately* normal with mean = $n\mu$ and variance $n\sigma^2$ (as if the distributions were normal)
  - The distribution of the sample mean $\overline{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$
  - Recall what we mean by "distribution of the sample mean"

5

Distribution of sample means



6

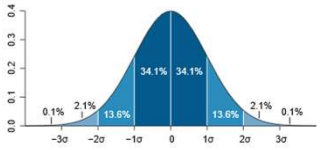
Example

- Observations $x_1, x_2, \ldots x_{100}$ [n=100] (or $x_1, x_2, \ldots x_{10}$) [n=10]
- Sample mean of observations: "estimate" denoted by $\overline{x}$
- If we want to ask questions like "what are the chances that the estimate will be …" we are abstracting and thinking about the random variable $\bar{X}$ (the estimator)
  - The CLT says that $\overline{\mathrm{X}} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$ regardless of where X_i came from (for large n)

89	68	92	74	76	65	77	83	75	87
85	64	79	77	96	80	70	85	80	80
82	81	86	71	90	87	71	72	62	78
77	90	83	81	73	80	78	81	81	75
82	88	79	79	94	82	66	78	74	72

7

CLT Example



- Your friend tells you that in 10000 successive independent unbiased coin tosses, he counted at least 5200 heads. Is (s)he serious?
  - This is a single sample from your friend with choice of n=10000
- Let $X_1, X_2, \ldots, X_n$ be the random variables indicating whether the coin toss was a success (head).
  - These are i.i.d. random variables whose sum is a random variable with:
  - Expected value $n\mu = 10000(0.5) = 5000$, $\sigma = \sqrt{(10000 * 0.5 * 1.5)} = 50$
  - By CLT, $\bar{X} \approx \mathcal{N}(5000, 2500)$ and the z-score corresponding to 5200 (with continuity correction) is (5199.5-5000)/50 = 3.99 (4 standard deviations)
- Your friend is joking (why?)

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8

Review

- We use both $\bar{x}$ and $\bar{X}$ to refer to a sample mean. For what purposes do we use the former and for what purposes do we use the latter?
- What is meant by "the sampling distribution of $\bar{X}$?
- How is the population mean of the sample average $\bar{X}$ related to the population mean of individual observations?
- How is the population standard deviation of $\bar{X}$ related to the population standard deviation of individual observations?
- What happens to the sampling distribution of $\bar{X}$ if the sample size is increased?

9

CLT for proportions

- 1000 American teenagers were interviewed. 27% reported having seen illegal drugs sold at school
- A single sample proportion is denoted by $\hat{p}$ (Here $\hat{p} = 0.27$)
- The corresponding random variable is $\hat{P}$
- The sample proportion $\hat{p}$ is an estimate of the true population parameter p from Bin(n,p)
- CLT: $\hat{P} \approx \mathcal{N}(p, \frac{p(1-p)}{n})$
  - In this example, B = Bin(1000, 0.27) itself is a (good) approximation, and the CLT is a further (good) approximation before

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10

CLT for proportions

- Your friend tells you that in 10000 successive independent unbiased coin tosses, he counted at least 5200 heads. Is (s)he serious?
- Let $\hat{p}$ be the sample proportion of heads
  - $E[\hat{p}] = p = 0.5,\ \ \mathrm{sd}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.25}{10000}} = 0.005$
- With continuity correction, zscore $= \frac{\hat{p}-p}{\sqrt{p(1-p)/n}} = \frac{0.51995-0.5}{0.005} = 3.99$
  - Your friend is joking

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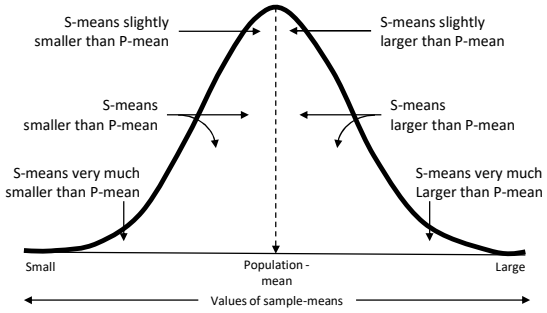
11

Review

You want	Continuity correction on count	On proportion
$P(X \geq k)$	$k - 0.5$	$(k - 0.5)/n$
$P(X \leq k)$	$k + 0.5$	$(k + 0.5)/n$
$P(X = k)$	$[\mathrm{k} - 0.5, \mathrm{k} + 0.5]$	$[(k - 0.5)/n, (k + 0.5)/n]$

12

Sample Mean vs Population Mean



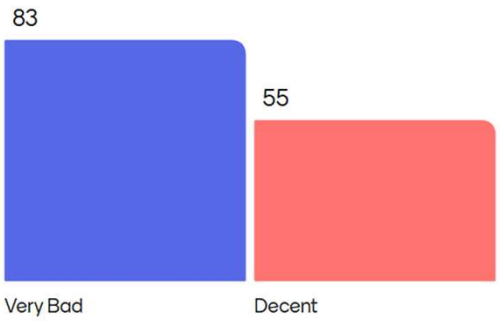
A single sample mean will give us a value that is different from the population mean

How bad is it?

13

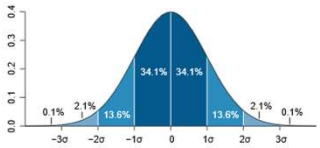
How bad is a single sample?

How bad is a single sample? **Can't say is not an option**



14

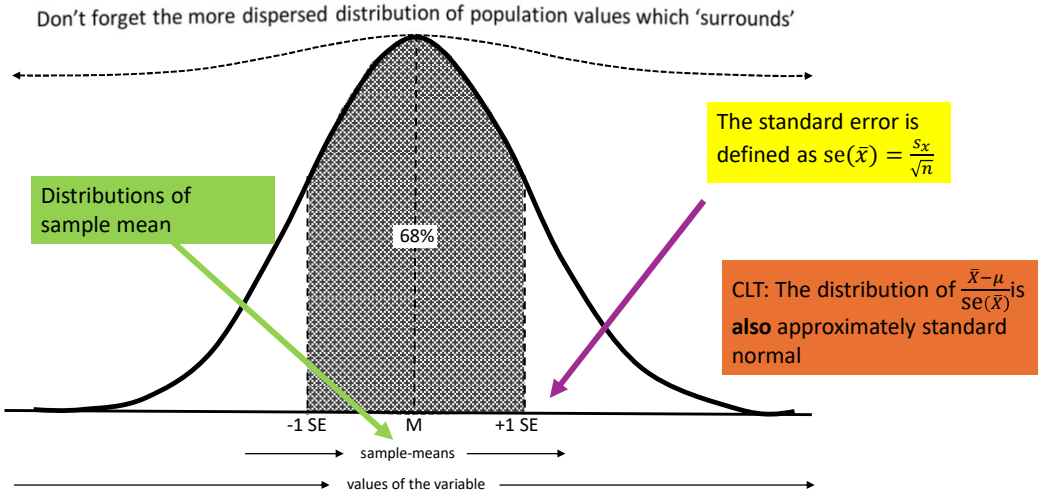
Logic of sampling



- The distance between $\bar{x}$ and μ is hardly bigger than $2\text{sd}(\bar{X})$ if we have a normal distribution for $\bar{X}$
  1. This could have been practically useless because nobody knows that their data is coming from a normal distribution
  2. No one knows the value of $\text{sd}(\bar{X})$
- However, CLT rescues us saying, yes, the population is normal in many cases, i.e., $\overline{\text{X}} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$ (note the approximation)
- We will address the second point shortly ("student distribution")
- Define: $\text{se}(\bar{x}) = \frac{s_x}{\sqrt{n}}$ where $s_x^2 = \frac{\sum_{i=1}^{n}(\text{x}_i - \overline{\text{x}})^2}{n-1}$
  - This is called the standard error of the sample mean

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Standard Error



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Logic of Sampling

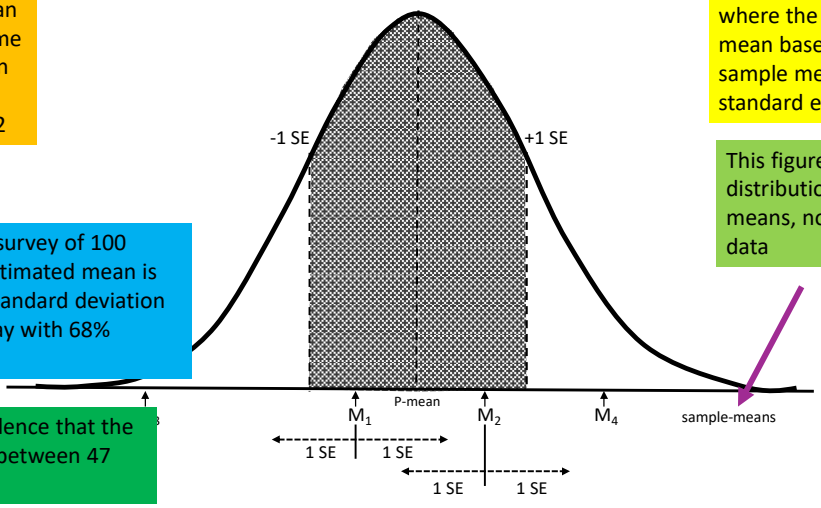
If the estimated mean is M2, 68% of the time our population mean will be within 1 standard error of M2

If we take a sample survey of 100 students, and the estimated mean is 50, and estimated standard deviation is 15, then we can say with 68% confidence that …

And with 95% confidence that the population mean is between 47 and 53

This figure tells us a very good idea about where the population mean based on a sample mean and standard error

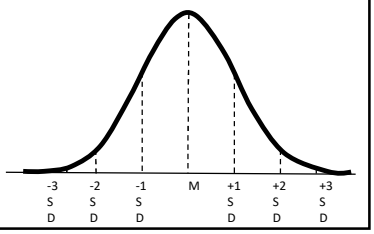
This figure is about the distribution of sample means, not the actual data



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From Samples To Population

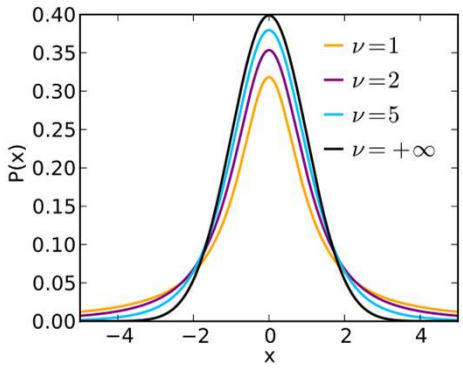
- Given a single pulse rate n=1, say 90, what can we say about the population distribution?
  - With n=1, our sample mean will not be accurate (but it won't be too off)
  - If we have a mean of a number of samples, we will have a sample standard deviation which will be different from the population standard deviation
- How wrong will we be? The size of the standard error of the mean depends on
  - The standard deviation within the sample
  - The size of the sample
  - The proportion of the population covered by the sample



20

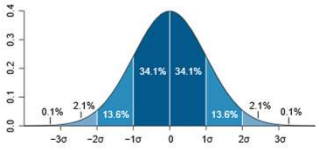
Agenda

- Last time
  - Informal approach to confidence intervals
  - Approximate distribution of the sample mean using CLT
  - Approximate distribution of the sample proportion using CLT
- Today
  - The “student” distribution



1

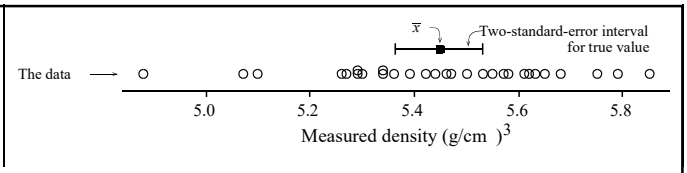
Logic of sampling



- The distance between $\bar{x}$ and μ is hardly bigger than $2\text{sd}(\bar{X})$ if we have a normal distribution for $\bar{X}$
  1. This could have been practically useless because nobody knows that their data is coming from a normal distribution
  2. No one knows the value of $\text{sd}(\bar{X})$
- However, CLT rescues us saying, yes,
  - The population is normal in many cases, i.e., $\bar{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$
  - Further, we have an even more convenient result $\bar{X} \approx \mathcal{N}(\mu, \text{se}(\bar{x}))$

2

Example



- In 1798 Cavendish measured the density of the earth
  - sample mean $\bar{x}$ = 5.447931 g/cm^3 and s_x = 0.2209457 g/cm^3
  - Therefore se($\bar{x}$) = 0.04102858
- From the theory, the true mean is $\bar{x} = 5.447931 \pm 2 \times 0.04102858$
- The number of digits of precision is unwarranted so we can say that Cavendish predicted true value between 5.37 and 5.53
- Current accepted value is ~ 5.517

5.50	5.61	4.88	5.07	5.26	5.55	5.36	5.29	5.58	5.65
5.57	5.53	5.62	5.29	5.44	5.34	5.79	5.10	5.27	5.39
5.42	5.47	5.63	5.34	5.46	5.30	5.75	5.68	5.85	

3

Review

- Why is the standard deviation of $\bar{X}$ not a useful measure of the precision of $\bar{X}$ as an estimator in practical applications?
- What measure of precision do we use in practice?
- How is se($\bar{x}$) related to the standard deviation of $\bar{X}$?
- When we use the formula se$(\bar{x}) = s_x/\sqrt{n}$, what is s_x and how do you obtain it?

4

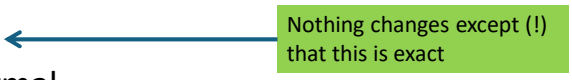
Beyond Approximation

- Although a good practical tool, there are two approximations
  - The CLT approximation for large n
  - The "se($\bar{x}$)" approximation for the unknown standard deviation
- Both approximations are problematic for small n
- If the population were known to be exactly normal, then we have exact results

5

Distribution of Sample mean

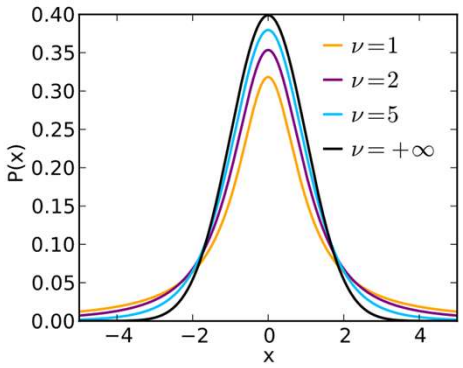
- For independent identical (iid) **normal** random variables, and $Y = X_1 + X_2 + \cdots X_n$ with $\bar{X} = Y/n$, $\mathbb{E}[X_i] = \mu$ and finite variance σ^2
  - $\mathbb{E}[\bar{X}] = \mu, \mathrm{Var}(\bar{X}) = \sigma^2/n$
  - $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$
  - $(\bar{X} - \mu)/(\sigma/\sqrt{n})$ is standard normal
- $\bar{x}$ is an estimate of the sample mean $\bar{X}$
  - $\bar{x} = \sum_i x_i \,/n$
  - This formula comes from the estimator $\bar{X}$ by simply observing the values



6

The student t-distribution

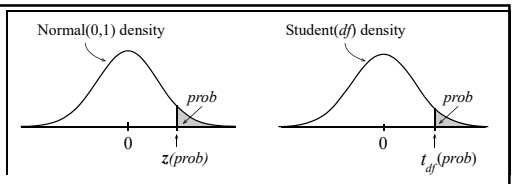
- If the population were known to be exactly normal, then we have the result $T_n = (\bar{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$
  - This is the student t-distribution with parameter "n-1"
- Here we have introduced
  - A new statistic S (it is a statistic not an estimator)
  - A new distribution T_n (df) where "df" stands for degrees of freedom



This is new, we get to use "S" instead of σ

7

Statistic for the variance



- S is easily recognizable
  - We define the random variable $S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$
  - S is the square root
- Thus, we have the practical result $(\bar{X} - \mu)/\mathrm{se}(\bar{X}) \sim t_{n-1}$
- For small n,
  - using t-values rather than z-values is robust even for non-normal distribution (empirical result)
  - For proportions, when the binomial assumption is valid, t is not needed
- For large n, t-distribution tends to the standard normal anyway
- In stat courses, tables for the t-distribution are available

8

Extracts from the Student's t-Distribution Table

prob — Desired *upper-tail prob*

Desired *df* →

df	.20	.15	.10	.05	.025	.01	.005	.001	.0005	.0001
6	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959	8.025
7	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408	7.063
8	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041	6.442
…	…	…	…	…	…	…	…	…	…	…
10	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587	5.694
…	…	…	…	…	…	…	…	…	…	…
15	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073	4.880
…	…	…	…	…	…	…	…	…	…	…
∞	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291	3.719

Some results

… and we just introduced T out of the blue!

9

#1: Expectation of S^2

- We can show that $E(S^2) = \sigma^2$
  - Given $S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$ we can show that $S^2 = \frac{\sum_{i=1}^{n} X_i^2 - n\overline{X}^2}{n-1}$
  - $E[(n-1)S^2] = E\left(\sum_{i=1}^{n} X_i^2 - n\overline{X}^2\right) = nE(X_i^2) - nE(\overline{X}^2)$
    - But $E(W^2) = \mathrm{Var}(W) + E(W)^2$
  - $E[(n-1)S^2] = n\,\mathrm{Var}(X_1) + nE(X_1)^2 - n\,\mathrm{Var}(\overline{X}) - nE(\overline{X})^2$
  - $E[(n-1)S^2] = n\sigma^2 + n\mu^2 - n(\sigma^2/n) - n\mu^2 = (n-1)\sigma^2$
- Note: this is not the expectation of S

10

$$\sum_{i=1}^{N}(X_i - \bar{X})^2 = \sum_{i=1}^{N}\left[(X_i - \mu) - (\bar{X} - \mu)\right]^2$$

$$= \sum_{i=1}^{N}(X_i - \mu)^2 - 2(\bar{X} - \mu)\sum_{i=1}^{N}(X_i - \mu) + \sum_{i=1}^{N}(\bar{X} - \mu)^2.$$

But

$$\sum_{i=1}^{N}(X_i - \mu) = \sum_{i=1}^{N} X_i - N\mu = N(\bar{X} - \mu),$$

and

$$\sum_{i=1}^{N}(\bar{X} - \mu)^2 = N(\bar{X} - \mu)^2.$$

Therefore

$$\boxed{\sum_{i=1}^{N}(X_i - \bar{X})^2 = \sum_{i=1}^{N}(X_i - \mu)^2 - N(\bar{X} - \mu)^2}$$

11

Bessel Correction

- If we had started with n in the denominator
- Then we have ended up with $nE(S^2) = (n-1)\,\sigma^2$
- In such the expectation would have a (n-1)/n factor
  - This is the Bessel's correction
  - For n = 2, we would have ended up with E(S^2) = ½ σ^2
  - This is exactly what we saw in the experiment on the first day of class

12

Experiment

Now focusing on the last two columns

All possible samples of size 2	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum(x-\bar{b})^2}{n-1}$	$\frac{\sum(x-\bar{b})^2}{n}$
0 0	(0+0)/2 = 0	$((0-0)^2+(0-0)^2)/1 = 0$	0
0 2	1	$((0-1)^2+(2-1)^2)/1 = 2$	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

- Considering samples of size two,
  - Dividing by n results in $E[\overline{s}^2] = \frac{12}{9} = \frac{4}{3}$
  - Division by n underestimates the true variance
  - Dividing by (n-1) results in $E[\overline{s}^2] = \frac{24}{9} = \frac{8}{3}$

13

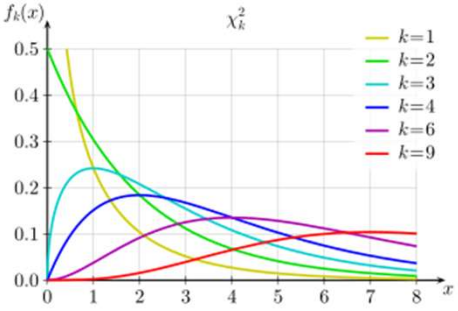
#2: Independence

- For independent identical (iid) **normal** random variables X_i with mean μ and variance σ^2
  - $\bar{X}$ is normal with mean μ and $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$
- Both $\bar{X}$ and S^2 are computed from the same data, yet they are independent, this is a surprising result (not proven here)
- Further
  - $\frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^{n}\left(\frac{X_i-\bar{X}}{\sigma}\right)^2$ which is the sum of standard normals from *iid* X_i
  - The rhs has a name! $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$
  - Thus distribution of the statistic $S^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}$

15

Chi-square distribution [1/3]

- If $Z_1, Z_2, \ldots, Z_n$ are independent *standard* normal then
  - $X = Z_1^2 + Z_2^2 + \cdots + Z_n^2$ is said to have a chi-square distribution with n degrees of freedom and is denoted as $X \sim \chi_n^2$
  - $f_X(x) = \frac{x^{n/2-1}e^{-x/2}}{2^{n/2}\Gamma(n/2)}$
  - $\Gamma(y) = \int_0^\infty x^{y-1}\, e^{-x} dx$
- Example (n=1)
  - $Z_1 \sim N(0,1), X = Z_1^2$
  - $f_X(x) = \frac{e^{-x/2}}{\sqrt{x}\sqrt{2\pi}}$



16

Chi-square distribution [2/3]

- In stat courses, tables for the chi-square distribution are available for different number of degrees of freedom, and for different values of the probability
- MGF: $\Phi_X(t) = (1-2t)^{-n/2}$ (t < ½)
- Additive property
  - If X1 and X2 are *independent* chi-square random variables with n1 and n2 degrees of freedom respectively, then X1 + X2 is also a chi-square random variable with n1 + n2 degrees of freedom

17

Chi-square distribution [3/3]

- Can we use the additive property?
  - $(n-1)S^2 = \sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2 = \sum_{i=1}^{n}(X_i-\mu)^2 - n\left(\overline{X}-\mu\right)^2$
  - $\frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\sigma^2} = \frac{\sum_{i=1}^{n}\left(X_i-\overline{X}\right)^2}{\sigma^2} + \frac{n\left(\overline{X}-\mu\right)^2}{\sigma^2}$
  - $\sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2 = \frac{\sum_{i=1}^{n}\left(X_i-\overline{X}\right)^2}{\sigma^2} + \left(\frac{\sqrt{n}\left(\overline{X}-\mu\right)}{\sigma}\right)^2$

Recognize standard normal

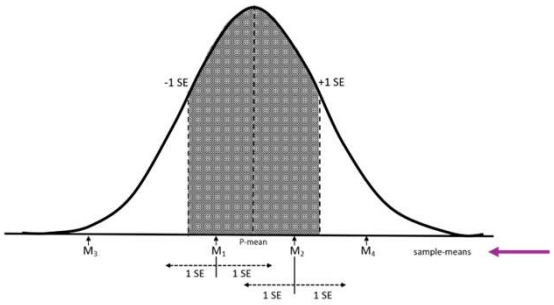
Recognize as χ_n^2

Recognize as $(n-1)/\sigma^2 S^2$

Independent

Agenda

- Last time
  - χ^2_n distribution
  - The S^2 statistic
  - Independence of S^2 and $\bar{X}$
- Today
  - Confidence interval



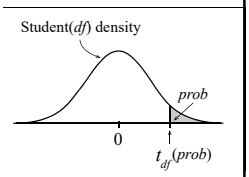
1

Approximate confidence interval

- Earlier: Cavendish conducted experiments seeking density of earth
  - sample mean of 29 observations $\bar{x}$ = 5.447931 g/cm^3 and s_x = 0.2209457 g/cm^3 and therefore se($\bar{x}$) = 0.04102858
  - True mean lies in the interval $\bar{x} = 5.447931 \pm 2 \times 0.04102858$
- Assumptions and problems
  - Experiments may be "biased"
  - Distribution of means is only approximately normal
  - Too many significant digits
- Coarse bound can be strengthened with the "student t-distribution"

3

Confidence interval with t-distribution



- Earlier: Cavendish conducted experiments with 29 values
  - $\bar{x}$ = 5.447931 s_x = 0.2209457 se($\bar{x}$) = 0.04102858
  - True mean lies in the interval $\bar{x} = 5.447931 \pm 2 \times 0.04102858$
- Coarse bound can be strengthened with the "student t-distribution"
  - $(\bar{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$
  - $-t_{\mathrm{df}_{0.0275}} \leq \Pr\left(\frac{\bar{x}-\mu}{\mathrm{se}(\bar{x})}\right) \leq t_{\mathrm{df}_{0.0275}}$
  - tinv(0.975,28) = 2.0484, so $\bar{x} \approx 5.45$, confidence interval = [5.36, 5.53]

Note 28, not 29

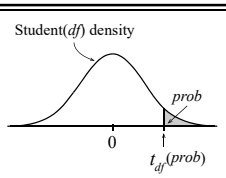
4

Significant digits

- Cavendish conducted experiments with 29 values
  - $\bar{x}$ = 5.447931 s_x = 0.2209457 $\mu \in 5.447931 \pm 2.0484 \times 0.04102858$
- Coarse bound can be strengthened with the "student t-distribution"
  - tinv(28, 0.975) = 2.0484, so $\bar{x} \approx 5.45$, confidence interval = [5.36, 5.53]
- Rules of thumb for significant digits
  - Intermediate calculations as accurate as possible
  - Estimate to be reported up to the first significant digit of se, se one more significant digit than the se, confidence interval maybe one more magnitude compared to the length of the interval (0.1681 in this case)

5

A 95% confidence interval



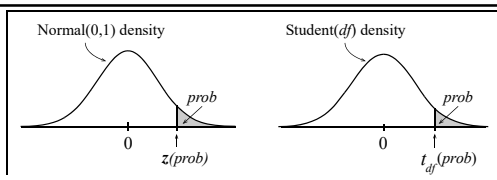
- When we estimate a population parameter θ using an estimate $\hat{\theta}$, the defining property of a 95% confidence interval is
  - "a type of interval that contains the true value of θ for 95% of samples"
- In our examples: a confidence interval is the $\hat{\theta} \pm t$ standard errors

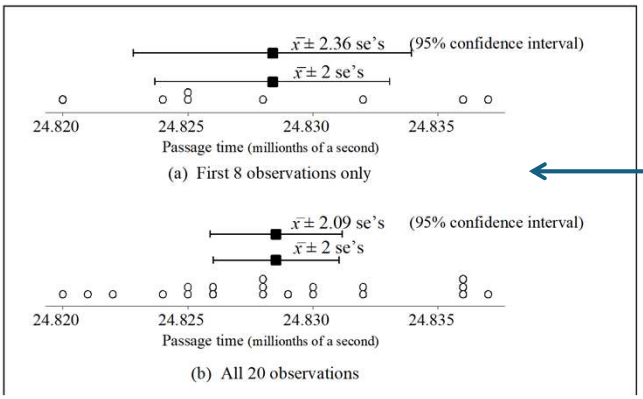
Value of the Multiplier, t, for a 95% CI

df :	7	8	9	10	11	12	13	14	15	16	17
t :	2.365	2.306	2.262	2.228	2.201	2.179	2.160	2.145	2.131	2.120	2.110
df :	18	19	20	25	30	35	40	45	50	60	∞
t :	2.101	2.093	2.086	2.060	2.042	2.030	2.021	2.014	2.009	2.000	1.960

6

Review example





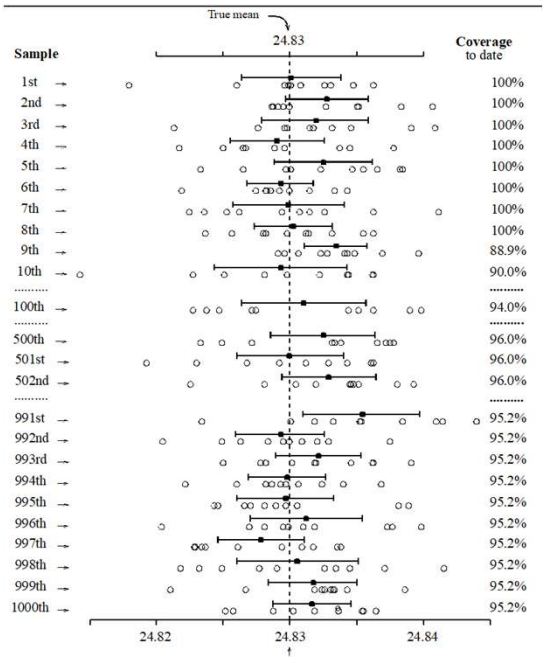
Newcomb's experiment to measure the speed of light in the year 1882

Time for a distance of 3721 m across the river Potomac

7

Example

- 1000 experiments independent samples of size 10 from $\mathcal{N}(\mu = 24.83, \sigma^2 = 0.005^2)$
- In each case the 95% confidence interval is overlaid
- What is your observation?
- Repeated sampling
- Many independent studies
- Planning a study: In my study I will capture the correct value 95% of the time



8

Review

- Effect of increasing the confidence level (df = ∞)

80% CI, $\bar{x} \pm 1.282\ \mathrm{se}(\bar{x})$

90% CI, $\bar{x} \pm 1.645\ \mathrm{se}(\bar{x})$

95% CI, $\bar{x} \pm 1.960\ \mathrm{se}(\bar{x})$

99% CI, $\bar{x} \pm 2.576\ \mathrm{se}(\bar{x})$

9

Effect of increasing sample size

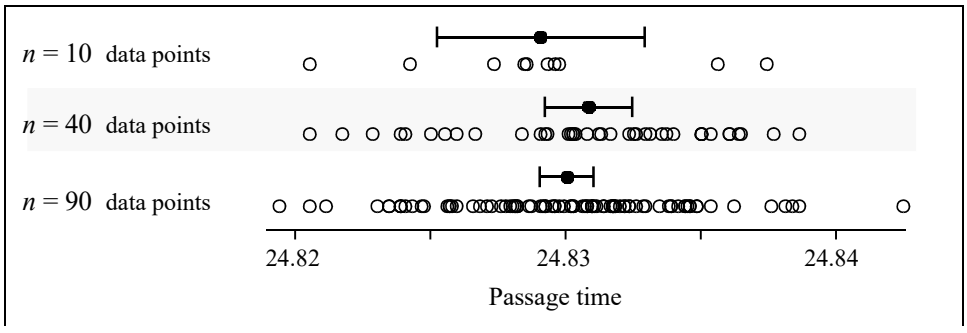
- If we have n observations and we want to increase the precision (say double it), how many more observations do we need?

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10

Effect of increasing sample size

- If we have n observations and we want to increase the precision (say double it), how many more observations do we need?



11

But what exactly is this t-distribution?

I can visualize the plot, I get df, I can use it to get the confidence interval, I can do significance digits but …

12

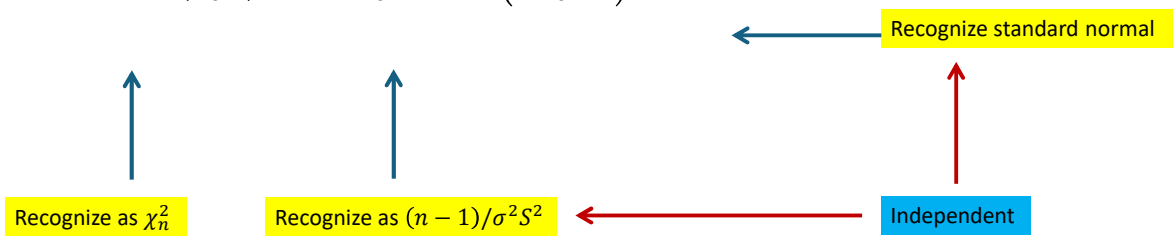
The t-distribution [1/2]

- Definition: If $Z \sim \mathcal{N}(0,1)$ and $V \sim \chi^2_{\mathrm{n}}$ are independent then
  - $T = \frac{Z}{\sqrt{V/\mathrm{n}}}$ is said to be a t-distribution with degrees of freedom *n*
- We have been using the result that if population were known to be normal, then $(\bar{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$
  - We must therefore show that the LHS is a ratio of a standard normal to the square root of an appropriate term
  - Divide the numerator and denominator by $\sigma/\sqrt{n}$
    - To recognize that the numerator is standard normal
    - And denominator is S/σ that is S^2/σ^2

13

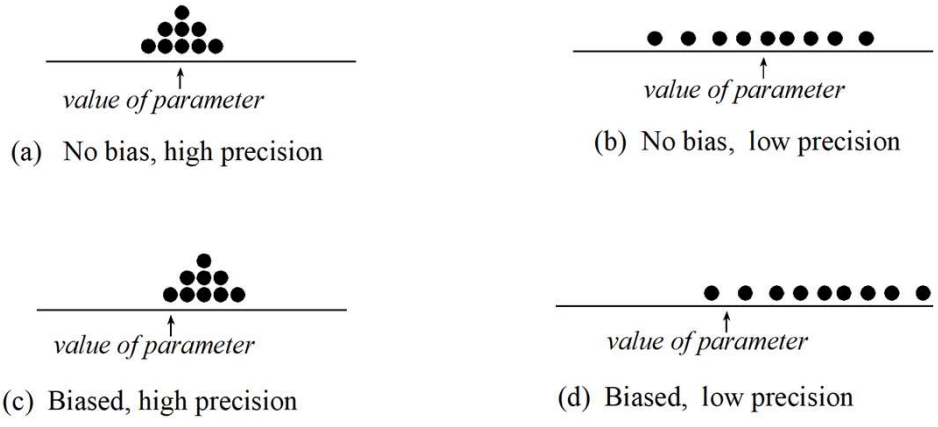
The t-distribution [2/2]

- But
  - $S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$ and $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$
  - And recall the independence from below to complete the result
  - $\sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2 = \frac{\sum_{i=1}^{n}(X_i-\overline{X})^2}{\sigma^2} + \left(\frac{\sqrt{n}(\overline{X}-\mu)}{\sigma}\right)^2$



14

Bias and precision



15

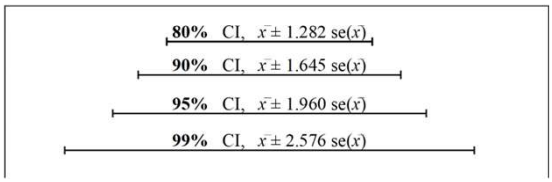
Unbiased Estimator

- The estimator $\bar{X}$ is an example of an unbiased estimator of the population mean because the expected value of $\bar{X}$ is μ
- The ***standard error*** of any estimate $\hat{\theta}$[denoted se($\hat{\theta}$)] estimates the variability of $\hat{\theta}$ values in repeated sampling and is a measure of the ***precision*** of $\hat{\theta}$

16

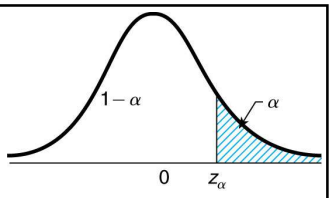
Agenda

- Last week
  - Is $\overline{\mathrm{X}} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$?
    - Large N, and Replacing σ by se$(\bar{x})$
  - χ^2_n distribution, The S^2 statistic
  - Independence of S^2 and $\bar{X}$
  - The t-distribution
  - $\mathbb{E}(S^2) = \sigma^2$
  - Confidence Interval
- Today
  - Hypothesis Testing

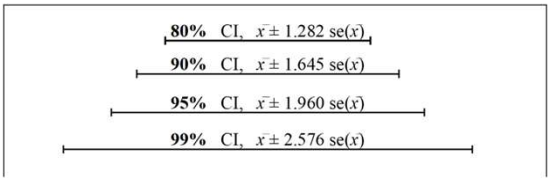


1

Context



- Given samples, report as accurate as possible a statistic
  - Make some inferences, but …
  - No measurement is accurate, so report error bounds



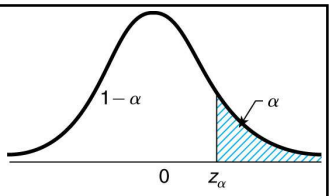
2

Recap for sample means

- With normal distributions, $(\bar{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$
- The standard error of the sample mean is $\text{se}(\bar{x}) = \frac{s_x}{\sqrt{n}}$
- 95% confidence interval: A type of interval that contains the true value of a parameter for 95% of samples
  - For small values of n, and with a normal distribution, and with $\alpha = 0.05$, $\bar{x} \pm t_{1-\alpha/2}(n-1)\text{se}(\bar{x})$ indicates the confidence interval
- Practice: We use the same expression for confidence interval for
  - Small n, and with approximately normal distribution,
  - With the CLT approximation (large n) as a conservative practice

3

Recap for sample proportions



- Small n: Cannot use the normal approximation for the Bin(n, p)
- Large n: $\hat{P} \approx \mathcal{N}(p, \frac{p(1-p)}{n})$ either via Stirling approximation or CLT applied to the Bernoulli
  - More formal: For $p \in (0,1)$ as $n \to \infty$, $\frac{\sqrt{n}(\hat{P}-p)}{\sqrt{p(1-p)}} \xrightarrow{d} \mathcal{N}(0,1)$
  - Large N: $np(1-p) > 10$ we use $\frac{(\hat{P}-p)}{\text{se}(\hat{p})} \xrightarrow{d} \mathcal{N}(0,1)$ where the standard error of the sample proportion is $\text{se}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
  - 95% confidence interval: With $\alpha = 0.05$, $\hat{p} \pm z_{1-\alpha/2}\text{se}(\hat{p})$
    - t-distribution not applicable; unlike the mean case, the variance is dependent on mean

4

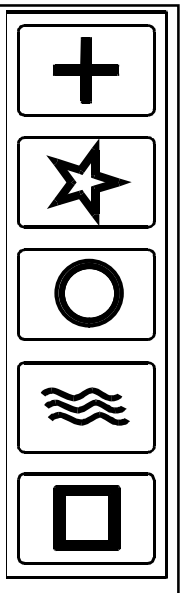
Hypothesis Testing

- Given samples, report as accurate as possible a statistic
  - Make some inferences, but …
  - No measurement is accurate, so report error bounds
  - We cannot conclusively prove a hypothesis
- Science is not a library of truth, it's a debugging process
  - Example: Mass is not absolute (Rest mass vs mass observed): $m_{\text{rel}} = \frac{m}{\sqrt{1-\frac{v^2}{c^2}}}$.
  - Being wrong is not a bug, it's a feature (engine?) of science

5

Example #1

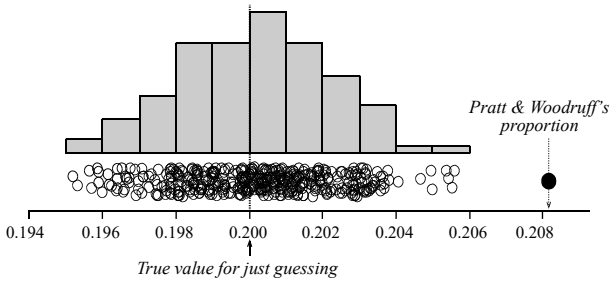
- In 1938, scientists showed 60,000 pictures
  - Students predict what is behind the card
  - If students guessed randomly there would be 12000 correct answers (proportion is 0.2)
  - … but actual result was 12489 (proportion is 0.208)
  - Is 20.8% being larger than 20% a "chance event" or
  - Hypothesis: Students can mind read correct answer in an MCQ



6

Disproving the hypothesis

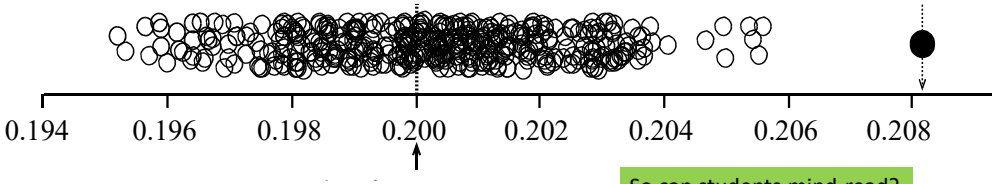
- Conduct computer simulation
  - of 400 experiments
  - Generate 60,000 possible "answers" on the back of the card
  - Guess with 0.2% probability one of the 5 cards



7

Hypothesis Testing

- We cannot "rule-in" a hypothesized value for a parameter
  - For example, no amount of computer experiments can conclusively say that the proportion is 0.201 vs 0.199
  - The graph would just shift to the right (or left)
- We can only determine whether there is evidence to rule out a hypothesized value

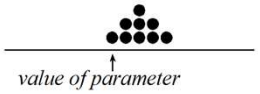


So can students mind-read?

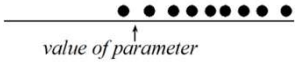
8

Example #2

- In 1798, Cavendish measured the density of earth
  - After 6 experiments, he had to change his equipment resulting in the data 5.36, 5.29, 5.58, 5.65, 5.57, 5.53, 5.62, 5.29, 5.44, 5.34, 5.79, 5.10, 5.27, 5.39, 5.42, 5.47, 5.63, 5.34, 5.46, 5.30, 5.75, 5.68, 5.85
  - n = 23, sample mean = 5.483, sample sd = 0.1904 (earlier, n=29)
    - Current accepted value 5.517
  - Hypothesis: Cavendish's experiments were biased



(c) Biased, high precision

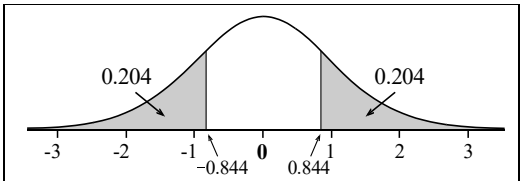


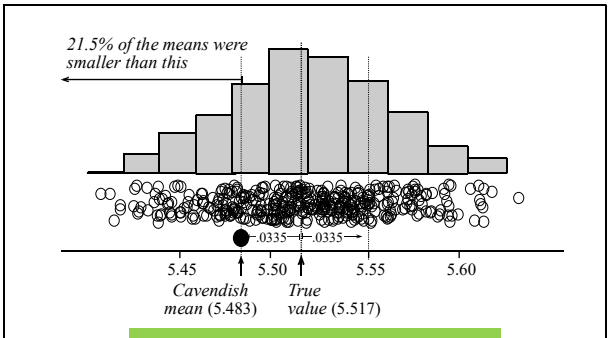
(d) Biased, low precision

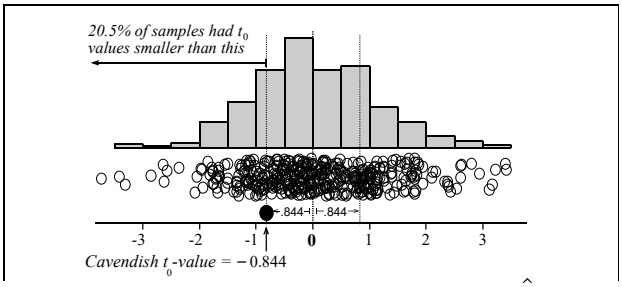
9

Disproving the hypothesis

- Computer simulation
  - 400 samples from an unbiased experiment







Was Cavendish experiment biased?

$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$

10

Review

- What intuitive criterion do we use to determine whether the hypothesized parameter value (p=0.2 in the ESP example, and μ = 5.517 was Cavendish) was *credible* in the light of the data?
- Why was it that μ = 5.517 may be considered credible but p=0.2 was *not credible*?
- Why do we switch to measuring distance between estimate and hypothesized parameter value in terms of standard errors?
  - What do t_0-values tell us?

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Hypothesis Testing

- In settling a research hypothesis, we turn skeptical and assume the research hypothesis is false
  - The null hypothesis is H_0 (or H_o)
  - The alternate is the research hypothesis H_1 or H_a
- Is there data to reject the null hypothesis?
  - Solution:
    - Compute the test statistic
      - Is this statistic "too far" from the null hypothesis?
    - If true, then reject the null hypothesis
    - If false, cannot reject the null hypothesis

$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$

Our goal is to reject the null hypothesis

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The t-test

- Using $\hat{\theta}$ to test $H_0: \theta = \theta_0$ versus H_a
  1. Calculate the test statistic t_0
  2. Calculate the p-value
  3. Interpret the p-value

P	%	Translation
> 0.12	-12%	No evidence against H0
0.1	-10%	Weak evidence against H0
0.05	-5%	Some evidence against H0
0.01	-1%	Strong evidence against H0
0.001	-0.10%	Very Strong evidence against H0

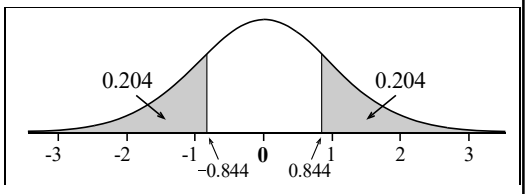
$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$

Alternative hypothesis	Evidence against H_0: $\theta > \theta_0$ provided by	P-value
H_1: $\theta > \theta_0$	too much bigger than θ_0 (i.e., $\hat{\Theta} - \theta_0$ too large)	P = pr(T ≥ t_0)
H_1: $\theta < \theta_0$	too much smaller than θ_0 (i.e., $\hat{\Theta} - \theta_0$ too negative)	P = pr(T ≤ t_0)
H_1: $\theta \neq \theta_0$	too far from θ_0 (i.e., $\lvert\hat{\Theta} - \theta_0\rvert$ too large)	P = 2 pr(T ≥ $\lvert t_0 \rvert$)
		where T ~ Student(df)

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Example #2

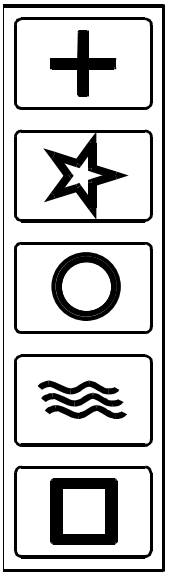
- In 1798, Cavendish measured the density of earth
  - After 6 experiments, he had to change his equipment resulting in the data 5.36, 5.29, 5.58, 5.65, 5.57, 5.53, 5.62, 5.29, 5.44, 5.34, 5.79, 5.10, 5.27, 5.39, 5.42, 5.47, 5.63, 5.34, 5.46, 5.30, 5.75, 5.68, 5.85
  - n = 23, sample mean = 5.483, sample sd = 0.1904 (Accepted value 5.517)
  - Research hypothesis: His last 23 experiments resulted in biased measurement
    - Null hypothesis: H_0: His experiments were not biased
    - Compute the t-statistic: -0.844
    - Compute the p-value: 0.204
    - Interpretation: Cannot refute H_o



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Example #1

- In 1938, scientists collected 60,000 pieces of data
  - If students guessed randomly
    - there would be 12000 correct answers, but actual result was 12489
  - Research hypothesis: Students mind read correct answer in an MCQ
  - Null hypothesis: proportion is 0.2
    - Compute the t-statistic: $\hat{p} = 0.20815$, p = 0.2, se=0.001657, t=4.92
    - Compute the p-value: < 10^{-6}
    - Interpret: Can refute the null hypothesis
    - Inference: Not pure guessing (maybe mind read?)



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Agenda

- Earlier
  - Hypothesis testing
  - Parameters and Statistic
- Today
  - Parameter estimation
  - MLE

Alternative hypothesis	Evidence against H_0: $\theta > \theta_0$ provided by	P-value
H_1: $\theta > \theta_0$	too much bigger than θ_0 (i.e., $\hat{\Theta} - \theta_0$ too large)	$P = pr(T \geq t_0)$
H_1: $\theta < \theta_0$	too much smaller than θ_0 (i.e., $\hat{\Theta} - \theta_0$ too negative)	$P = pr(T \leq t_0)$
H_1: $\theta \neq \theta_0$	too far from θ_0 (i.e., $\lvert\hat{\Theta} - \theta_0\rvert$ too large)	$P = 2\ pr(T \geq \lvert t_0 \rvert)$
		where T ~ Student(df)

1

Parameter and estimates

- A ***parameter*** is a numerical characteristic of a population or distribution. It often is used to "index" the distribution
- A **statistic** is a random variable and a function of the sample random variables $X_1,...,X_n$
- **Estimator**: A statistic $T(X_1,...,X_n)$ that is specifically being used as a guess for an unknown parameter θ
  - Example: The estimator $\bar{X}$
- After you actually observe, an ***estimate*** $\bar{x}$ is a realized quantity calculated from the data to estimate an unknown parameter
  - So we have done parameter estimation already! What's new?

2

Parameters

- Bernoulli: $\theta = p$ (probability of success)
- Poisson: $\theta = \lambda$ (rate parameter)
- Uniform: θ is a vector of the upper and lower bounds
- Normal: θ is a vector of μ and σ^2
- More complex distributions such as the beta or the pareto distribution might well have more parameters
- Goal: How to find the parameters

3

Context and workflow

4

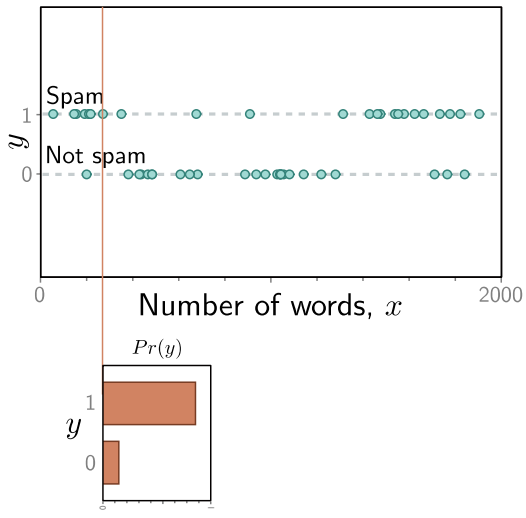
Context

- Step 1: “Real” business problems
  - Estimating the probability that at least two out of four servers will be alive next day
  - Is this email spam?
  - The probability that stock price will rise by 10% in the next week
  - The expected number of typos seen by a teacher in the next project report

5

Context

- Step 2: Model the problem. Choose a functional form of the uncertainty.
  - Is this email spam?
  - Choose Bernoulli as the model, and find out “success” probability



6

Context

- Step 1: Real business problems
  - Estimating the probability that at least two out of four servers will be alive next day
  - The probability that stock price will rise by 10% in the next week
  - The expected number of typos in project reports seen by a teacher in school
- Step 2: Model the problem. Choose a functional form of the uncertainty.
  - Bernoulli? Binomial? Gaussian? Poisson?

7

Context

- Step 3: Collect training samples
  - Number of words in an email message
  - Sample server failure data observed over (say) n=3 days
  - Stock price change over the last 10 days
  - Number of typos in past reports
- Step 4: Estimate the unknown parameters using the training sample

8

Review

- In the Cavendish experiment
  - What was the “real problem”
  - What was the model employed?
  - What kind of data collected?
  - How were the parameters reported? (Hint: use the word “interval” in your answer

9

Fitting parameters to model

10

Components of Solution

- Given: a density $f(x, \theta)$ with parameters θ
- Given: sample: $D = \{x_1, x_2, \ldots, x_I\}$
  - The i-th sample is a random variable X_i assumed to be iid as per $f(x, \theta)$
- Find θ
  - Since D is a finite sample, we cannot really know the actual θ. Best we can do is obtain an estimate of θ.
  - We will denote the estimate as $\hat{\theta}$

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Three methods

- Maximum likelihood
  - Sample D is all you have got
  - We get a *point* estimator $\hat{\Theta}$
- Maximum a posteriori and Bayesian approach
  - We also have some "prior" information

12

Maximum Likelihood (MLE)

- As the name suggests: find the parameters under which the data $x_{1\ldots I}$ are most likely
- When viewed as a function of Θ $p(D|\Theta)$ is called the likelihood function and it is not necessarily a probability
- When viewed as a function of x_i, the conditional probability $p(D|\Theta)$ is a probability distribution that sums to 1

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \Pr(\mathbf{x}_{1\ldots I}|\boldsymbol{\theta})$$
$$= \arg\max_{\boldsymbol{\theta}} \left[\prod_{i=1}^{I} \Pr(\mathbf{x}_i|\boldsymbol{\theta})\right]$$

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Bayes' Rule Terminology

Likelihood – propensity for observing a certain value of x given a certain value of y

Prior – what we know about y before seeing x

$$\Pr(y|x) = \frac{\Pr(x|y)\Pr(y)}{\int \Pr(x|y)\Pr(y)\,dy}$$

Posterior – what we know about y after seeing x

Evidence – a constant to ensure that the left-hand side is a valid distribution

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Maximum a posteriori (MAP) [1/2]

- As the name suggests we find the parameters which maximize the posterior probability $\Pr(\theta|x_{1\ldots I})$

$$\begin{aligned}\hat{\theta} &= \arg\max_{\theta} \Pr(\theta|\mathbf{x}_{1\ldots I}) \\ &= \arg\max_{\boldsymbol{\theta}} \left[\frac{\Pr(\mathbf{x}_{1\ldots I}|\boldsymbol{\theta})\Pr(\boldsymbol{\theta})}{\Pr(\mathbf{x}_{1\ldots I})}\right] \\ &= \arg\max_{\boldsymbol{\theta}} \left[\frac{\prod_{i=1}^{I}\Pr(\mathbf{x}_i|\boldsymbol{\theta})\Pr(\boldsymbol{\theta})}{\Pr(\mathbf{x}_{1\ldots I})}\right]\end{aligned}$$

Again we have assumed that data was independent

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Maximum a posteriori (MAP) [2/2]

Maximize the posterior probability $\Pr(\theta|x_{1\ldots I})$

$$\hat{\theta} = \arg\max_{\boldsymbol{\theta}} \left[\frac{\prod_{i=1}^{I}\Pr(\mathbf{x}_i|\boldsymbol{\theta})\Pr(\boldsymbol{\theta})}{\Pr(\mathbf{x}_{1\ldots I})}\right]$$

Point estimate

Since the denominator doesn't depend on the parameters we can instead maximize

$$\hat{\theta} = \arg\max_{\theta} \left[\prod_{i=1}^{I}\Pr(\mathbf{x}_i \mid \theta)\Pr(\theta)\right]$$

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Bayesian Approach [1/2]

Compute the posterior *distribution* over possible parameter values using Bayes' rule:

$$\Pr(\boldsymbol{\theta} \mid \mathbf{x}_{1\ldots I}) = \frac{\prod_{i=1}^{I} \Pr(\mathbf{x}_i \mid \boldsymbol{\theta}) \Pr(\boldsymbol{\theta})}{\Pr(\mathbf{x}_{1\ldots I})}$$

Not a point estimate

Principle: why pick one set of parameters? There are many values that could have explained the data. Try to capture all possibilities

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Predictive densities for 3 methods

Maximum likelihood:
Evaluate new data point $\boldsymbol{x}^*$ under probability distribution with ML parameters $\Pr(\mathrm{x}^* | \widehat{\boldsymbol{\theta}})$

Maximum a posteriori:
Evaluate new data point $\boldsymbol{x}^*$ under probability distribution with MAP parameters $\Pr(\mathrm{x}^* | \widehat{\boldsymbol{\theta}})$

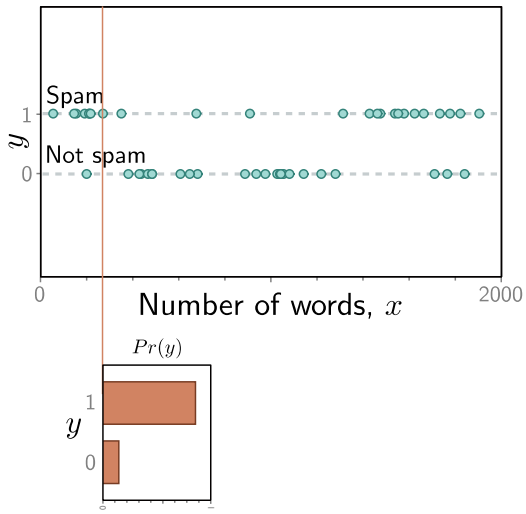
Bayesian:
Calculate weighted sum of predictions from all possible values of parameters

$$\Pr(\mathbf{x}^* \mid \mathbf{x}_{1\ldots I}) = \int \Pr(\mathbf{x}^* \mid \boldsymbol{\theta}) \Pr(\boldsymbol{\theta} \mid \mathbf{x}_{1\ldots I}) \, d\boldsymbol{\theta}$$

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Agenda

- Earlier
  - Typical workflow of a data scientist (Four steps)
  - Estimator and estimate
  - Three methods to get an estimate
  - Point estimate
  - Likelihood vs probability
- Today
  - MLE for normal
  - Linear regression



2

Context and workflow

3

Problem Definition

- Given some data samples $D = \{x_1, x_2, \ldots, x_I\}$
- Given some intended distribution $\Pr(\mathbf{x}_{1\ldots I}|\boldsymbol{\theta})$
  - Bernoulli: $\theta = p$ (probability of success)
  - Poisson: $\theta = \lambda$ (rate parameter)
  - Uniform: θ is a vector of the upper and lower bounds
  - Normal: θ is a vector of μ and σ^2
- Find the value of the parameter that "optimally" fits the distribution assuming $\mathrm{X}_1, \mathrm{X}_2, \ldots, \mathrm{X}_n$ are mutually independent (given θ)

4

Optimal Fitting parameters to model

Three methods

5

Maximum Likelihood (MLE)

- As the name suggests: find the parameters under which the data $x_{1\ldots I}$ are most likely
- When viewed as a function of Θ $p(D|\Theta)$ is called the likelihood function and it is not necessarily a probability
- When viewed as a function of X_i, the conditional probability $p(D|\Theta)$ is a probability distribution that sums to one

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \Pr(\mathbf{x}_{1\ldots I}|\boldsymbol{\theta})$$
$$= \arg\max_{\boldsymbol{\theta}} \left[\prod_{i=1}^{I} \Pr(\mathbf{x}_i|\boldsymbol{\theta})\right]$$

6

MLE Derivation

7

Fitting normal distribution: ML

- As the name suggests we find the parameters under which the data $x_{1\ldots I}$ is most likely

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}}\Big[\Pr(\mathbf{x}_{1\ldots I} \mid \boldsymbol{\theta})\Big]$$
$$= \arg\max_{\theta}\left[\prod_{i=1}^{I}\Pr(\mathbf{x}_i \mid \theta)\right]$$

Likelihood given by pdf

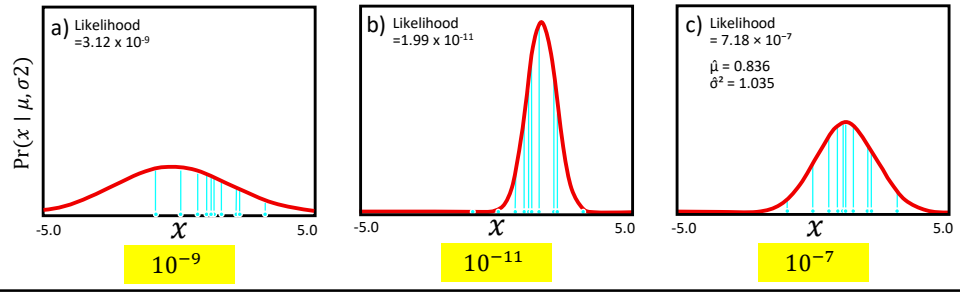
$$\Pr(x \mid \mu, \sigma^2) = \mathrm{Norm}_x[\mu, \sigma^2] = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-0.5\frac{(x-\mu)^2}{\sigma^2}\right]$$

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Fitting normal distribution: ML

$$\Pr(x_{1\ldots I} \mid \mu, \sigma^2) = \prod_{i=1}^{I}\Pr(x_i \mid \mu, \sigma^2)$$
$$= \prod_{i=1}^{I}\mathrm{Norm}_{x_i}[\mu, \sigma^2]$$
$$= \frac{1}{(2\pi\sigma^2)^{I/2}}\exp\left[-0.5\sum_{i=1}^{I}\frac{(x_i-\mu)^2}{\sigma^2}\right]$$

Ten data points



9

Fitting normal distribution: ML

Algebraically: $\hat{\mu}, \hat{\sigma}^2 = \arg\max_{\mu,\sigma^2} \left[\Pr(x_{1\ldots I} \mid \mu, \sigma^2) \right]$

where: $\Pr(x \mid \mu, \sigma^2) = \text{Norm}_x[\mu, \sigma^2]$

$$\Pr(x_{1\ldots I} \mid \mu, \sigma^2) = \prod_{i=1}^{I} \text{Norm}_{x_i}[\mu, \sigma^2] \quad = \frac{1}{(2\pi\sigma^2)^{I/2}} \exp\left[-0.5 \sum_{i=1}^{I} \frac{(x_i - \mu)^2}{\sigma^2} \right]$$

or alternatively, we can maximize the **logarithm**

$$\begin{aligned}\hat{\mu}, \hat{\sigma}^2 &= \arg\max_{\mu,\sigma^2} \left[\sum_{i=1}^{I} \log\left[\text{Norm}_{x_i}[\mu, \sigma^2] \right] \right] \\ &= \arg\max_{\mu,\sigma^2} \left[-0.5I\log[2\pi] - 0.5I\log\sigma^2 - 0.5\sum_{i=1}^{I} \frac{(x_i - \mu)^2}{\sigma^2} \right]\end{aligned}$$

10

Fitting normal distribution

$$\begin{aligned}\hat{\mu}, \hat{\sigma}^2 &= \arg\max_{\mu,\sigma^2} \left[\sum_{i=1}^{I} \log\left[\text{Norm}_{x_i}[\mu, \sigma^2] \right] \right] \\ &= \arg\max_{\mu,\sigma^2} \left[-0.5I\log(2\pi) - 0.5I\log\sigma^2 - 0.5\sum_{i=1}^{I} \frac{(x_i - \mu)^2}{\sigma^2} \right]\end{aligned}$$

$$\begin{aligned}\frac{\partial L}{\partial \mu} &= \sum_{i=1}^{I} \frac{(x_i - \mu)}{\sigma^2} \\ &= \frac{\sum_{i=1}^{I} x_i}{\sigma^2} - \frac{I\mu}{\sigma^2} = 0\end{aligned}$$

Solution: $\hat{\mu} = \frac{\sum_{i=1}^{I} x_i}{I}$

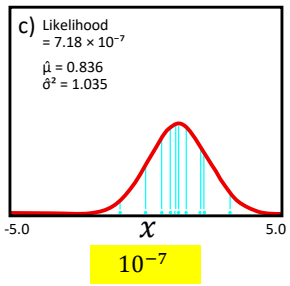
11

Fitting normal distribution: ML

Maximum likelihood solution:

$$\hat{\mu} = \frac{\sum_{i=1}^{I} x_i}{I}$$

$$\hat{\sigma}^2 = \sum_{i=1}^{I} \frac{(x_i - \hat{\mu})^2}{I}$$



Looks familiar?!

12

MLE

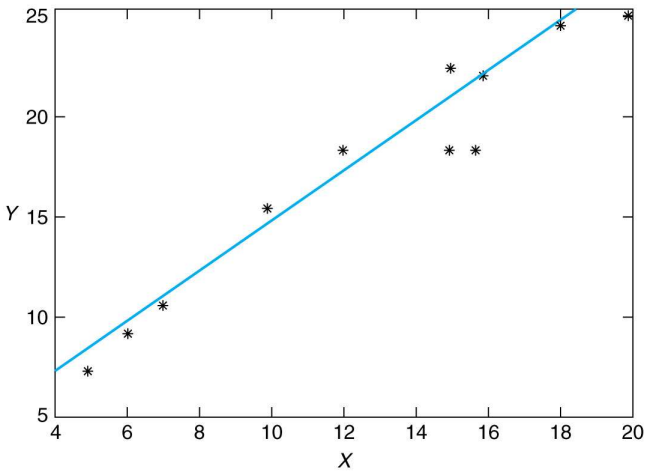
- Bernoulli: $\hat{p}_{\mathrm{ML}} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$
- Poisson: $\hat{\lambda}_{\mathrm{ML}} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$
- Uniform: $\hat{v} = X_{(1)} = \min_i X_i\,, \qquad \hat{\theta} = X_{(n)} = \max_i X_i$

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Linear Regression

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Linear Regression

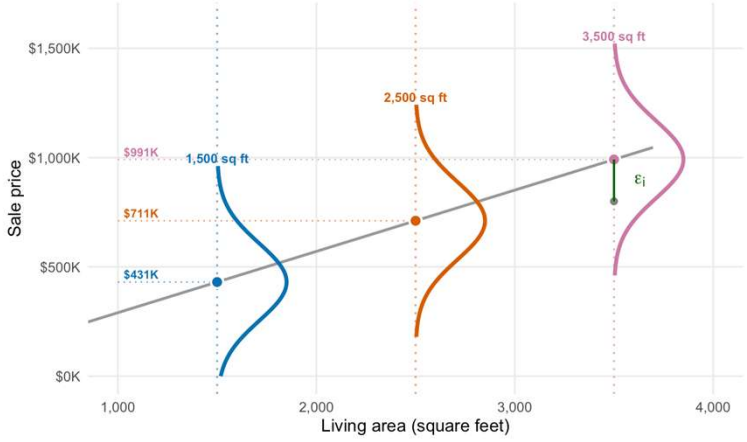


Most common case:
k=1

15

15

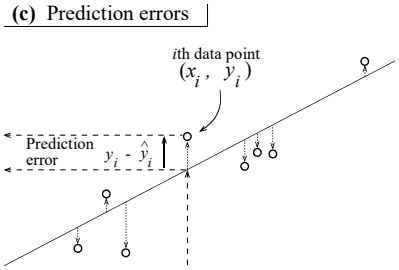
Linear Regression

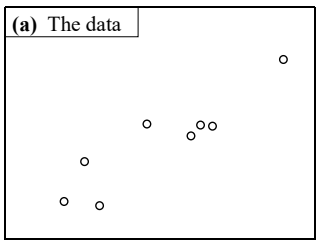


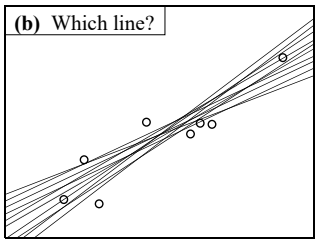
16

Fitting points to a line

- Goal is to get the "best" possible line
  - Estimated line: A + Bx
  - Minimize sum of squared vertical residuals





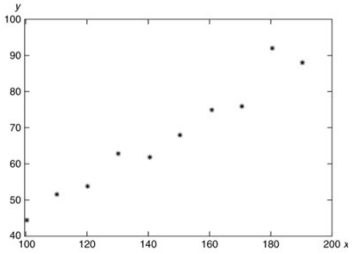


One (set of) outputs, one (set of) inputs

3

Fitting points to a line

- Goal is to get the "best" possible line
  - Estimated regression line: A + Bx
  - Minimize sum of squared vertical residuals
- In this form, there is no mention of uncertainty
- Solution via linear algebra: minimize norm($Up - v$)
  - $U_{n\times 2}$: [**1** x_i]
  - $p_{2}x_{1}$ = $(A,B)^T$ (estimators)
  - $v_{n\times 1}$: Y_i

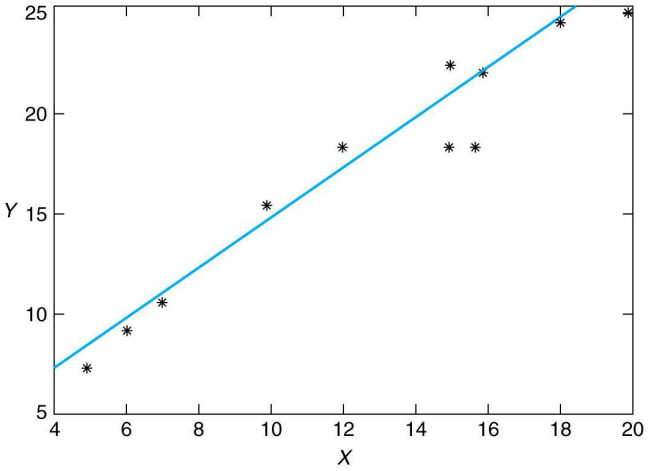


One output, one input

4

4

Fitting points to a line



$$B = \frac{\sum_{i=1}^{n}(x_i - \bar{x})\,(Y_i - \bar{Y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

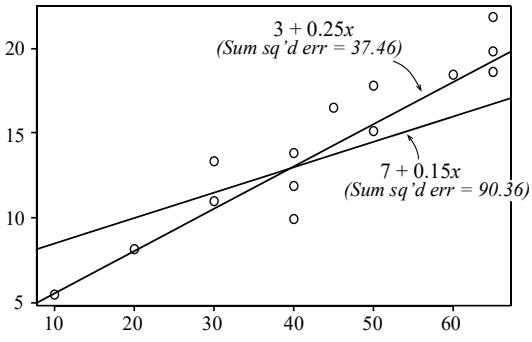
$$A = \bar{Y} - B\bar{x}$$

Proof on board

7

7

Review



Prediction Errors

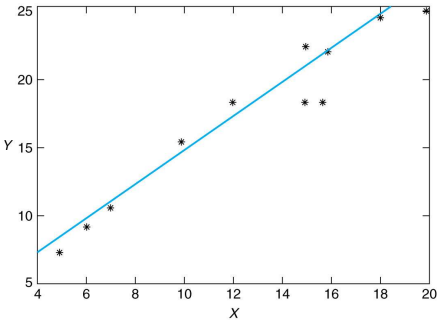
		3 + 0.25x		7 + 0.15x	
x	y	$\hat{y}$	$y-\hat{y}$	$\hat{y}$	$y-\hat{y}$
40	9.90	13.00	-3.10	13.00	-3.10
50	17.80	15.50	2.30	14.50	3.30
60	18.40	18.00	0.40	16.00	2.40
45	16.50	14.25	2.25	13.75	2.75
40	11.90	13.00	-1.10	13.00	-1.10
10	5.50	5.50	0.00	8.50	-3.00
30	11.00	10.50	0.50	11.50	-0.50
20	8.10	8.00	0.10	10.00	-1.90
50	15.10	15.50	-0.40	14.50	0.60
30	13.30	10.50	2.80	11.50	1.80
65	21.80	19.25	2.55	16.75	5.05
40	13.80	13.00	0.80	13.00	0.80
65	18.60	19.25	-0.65	16.75	1.85
65	19.80	19.25	0.55	16.75	3.05
	Sum of squared errors		37.46		90.36

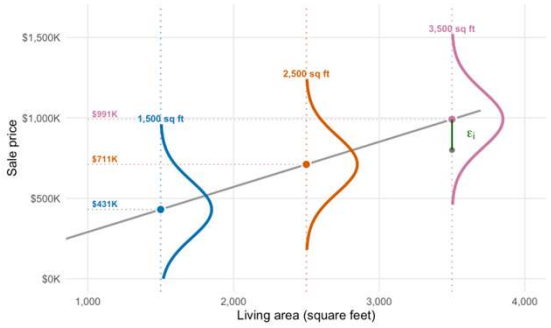
8

Linear Regression

- Consider the linear model
  - $Y_i = \beta x_i + \alpha + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$

β, α, x_i are assumed to be constant. We don't know β and α





$Y_i \sim \mathcal{N}(\beta x_i + \alpha, \sigma^2)$

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Least-squares fitting as MLE [1/2]

- Consider the linear model
  - $Y_i = \beta x_i + \alpha + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$
  - $p(y_i \mid x_i, \beta, \alpha) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(y_i - \beta x_i - \alpha)^2}{2\sigma^2}\right]$
  - $\ell(\beta, \alpha, \sigma) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta x_i - \alpha)^2 - n\log\left(\sigma\sqrt{2\pi}\right)$
- Maximizing likelihood gives the least-squares rule; applied to Y_i, it results in the normal equations

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \Pr(\mathbf{x}_{1\ldots I}|\boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta}} \left[\prod_{i=1}^{I} \Pr(\mathbf{x}_i|\boldsymbol{\theta})\right]$$

normal equations

$$\sum_{i=1}^{n} x_i Y_i = B\sum_{i=1}^{n} x_i^2 + A\sum_{i=1}^{n} x_i,$$

$$\sum_{i=1}^{n} Y_i = B\sum_{i=1}^{n} x_i + nA$$

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Least-squares fitting as MLE [2/2]

- Consider the linear model: $Y_i = \beta x_i + \alpha + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$
- Solving the "normal" equations results in
- $\boxed{B = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}, \qquad \boxed{A = \bar{Y} - B\bar{x}},$
- where $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i\,, \qquad \bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$

$$b_1 := \hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{(\sum_{i=1}^{n} x_i y_i) - n\bar{x}\bar{y}}{(\sum_{i=1}^{n} x_i^2) - n\bar{x}^2} \left(= r\frac{s_Y}{s_X}\right) \qquad b_0 := \hat{\beta}_0 = \bar{y} - b_1\bar{x}$$

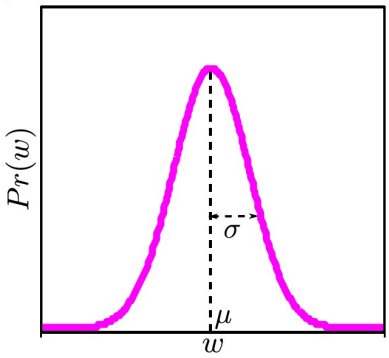
Formula: observed values of our intercept and slope estimators A,B.

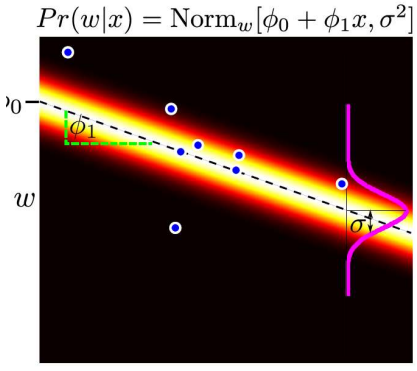
$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

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Linear Regression Review

)





Most common case:
k=1

$$Pr(w_i|\mathbf{x}_i, \boldsymbol{\theta}) = \text{Norm}_{w_i}\left[\phi_0 + \boldsymbol{\phi}^T\mathbf{x}_i, \sigma^2\right]$$

Image notation: w is our response Y; ϕ_0=α, ϕ_1=β; ϕ is the coefficient vector.

12

12

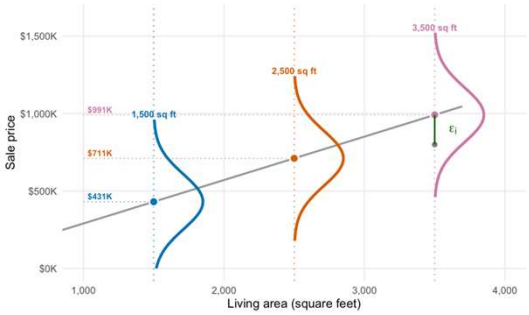
Regression Terminology

- Y is called output (response, dependent) variable
  - Also outcome, target, label, explained, predicted, regressand
- $(x_1, \dots, x_k)$ are called input or covariate or independent variables
  - Also predictor, feature, regressor, explanatory, attribute, factor, design, …
- We want to estimate the conditional density: $f(Y|\, x_1, \dots, x_k)$
  - $E = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_\mathrm{n}, y_\mathrm{n})\} = \{(x_{i_1}, x_{i_2}, \dots x_{i_k}, y_i) : i = 1, \dots, \mathrm{n}\}$

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Agenda

- Earlier
  - Simple Line Fitting
  - Linear algebra approach
  - MLE approach
- Today
  - Distribution of the estimators
  - Unknown variance
  - Multiple Linear regression



$$Y_i \sim \mathcal{N}(\alpha + \beta x_i, \sigma^2)$$

2

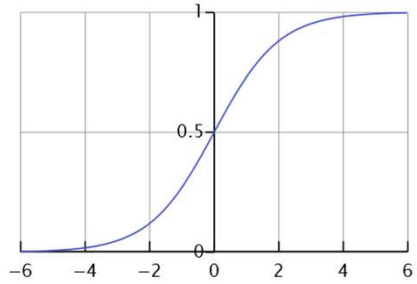
Regression asides

OLS, Logistic, polynomial, multiple

3

Logistic Regression

- Consider $\eta = \beta_1 x_1 + \cdots + \beta_k x_k + \alpha$
- Normal linear regression: $\mathbb{E}[Y|x] = \eta$
- Logistic regression: Probability of Y =1 where $\Pr(Y = 1|x) = \sigma(\eta)$
  - $\sigma(\eta) = \frac{1}{1+e^{-\eta}}$
  - Parameter estimates use iterative methods.



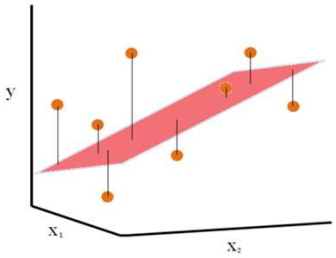
4

Polynomial Regression

- This is still linear in the parameters
- $Y = \beta_1 x + \beta_2 x^2 + \cdots + \beta_r x^r + \alpha + \epsilon$
- OLS treats x, x^2, …, x^r as the predictor columns.

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Multiple Linear Regression



- $Y \mid x_1, \ldots, x_k \sim \mathcal{N}(\mu_x, \sigma^2)$
- $\mu_x = \beta_0 + \sum_{j=1}^{k} \beta_j x_j$
- $Y = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \varepsilon$
- Training data: $\{(x_{i1}, x_{i2}, \ldots x_{ik}, y_i): i = 1 \ldots n\}$
- $B = \arg\min_\beta \sum_{i=1}^{n} \left(Y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2$
- $Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_k \end{bmatrix}$
- $Y = X\beta + \varepsilon$

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Review: Parameter and estimates

- A ***parameter*** is a numerical characteristic of a population or distribution. It often is used to "index" the distribution
- A **statistic** is a random variable and a function of the sample random variables $X_1,\ldots,X_n$
- **Estimator**: A statistic $T(X_1,\ldots,X_n)$ that is specifically being used as a guess for an unknown parameter θ
  - Example: The estimator $\bar{X}$
- After you actually observe, an ***estimate*** $\bar{x}$ is a realized quantity calculated from the data to estimate an unknown parameter

Oct06

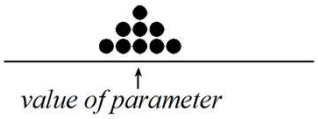
7

Understanding the derived parameters

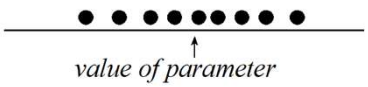
A look at bias, variance and mean-squared error of an estimator

8

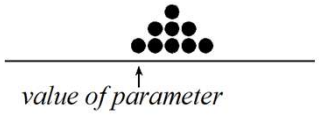
Review: Bias and precision



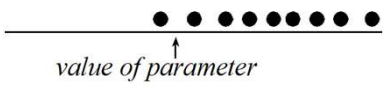
(a) No bias, high precision



(b) No bias, low precision



(c) Biased, high precision



(d) Biased, low precision

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Estimators are random variables

- The sample mean estimator is a random variable
- For independent and identically distributed (iid) random variables with a population mean μ, and finite variance σ^2
  - Distribution only for normal: $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$
  - Sample mean: *unbiased* estimator of population mean because $\mathbb{E}[\bar{X}] = \mu$
  - We also defined the estimator $S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$
    - Distribution only for normal: $S^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}$
  - For i.i.d. samples with finite variance: Unbiased estimator of the variance because $\mathbb{E}[S^2] = \sigma^2$

Note the denominator

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ML Estimators

- Maximum likelihood estimation requires a specified statistical model.
  - The model supplies the likelihood used to estimate its parameters
  - If you have no parametric model in mind, then the MLE recipe is probably irrelevant
- The Maximum Likelihood estimator is also a random variable
  - Why? Because its value is a function of the samples from some underlying distribution.
  - The ML estimator has its own sampling distribution, mean and variance.
  - An ML estimate is the observed value of the ML estimator.

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Estimator bias

- The expected value of the estimator, i.e. E($\hat{\theta}$) may not be equal to the true parameter θ, i.e., it may be biased.
  - Define: Bias of the estimator: $E(\hat{\theta}) - \theta$
- Earlier: when given samples are modeled as coming from a normal
  - ML estimator for the mean of a Gaussian is unbiased
  - ML estimator for the variance is biased
  - For large n, the bias is not a major issue

$$\hat{\mu} = \frac{\sum_{i=1}^{I} x_i}{I}$$

Observed μ̂=x̄, with I=n (our usual notation). The random mean estimator is X̄.

Note the denominator

$$\hat{\sigma}^2 = \sum_{i=1}^{I} \frac{(x_i - \hat{\mu})^2}{I}$$

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Estimator variance

- Merely knowing the estimator bias does not give us a complete idea of the quality of the estimator.
  - What if different samples of X gave wildly different estimates of θ for the same estimator?
- The variance of an estimator is (as expected!):

$$\mathrm{Var}(\hat{\theta}) = \mathbb{E}\left[\left(\hat{\theta} - \mathbb{E}[\hat{\theta}]\right)^2\right]$$

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Evaluating a point estimator

- Let X= $(X_1, X_2, \ldots, X_n)$ be a set of n i.i.d. random variables from a given distribution with parameter θ.
- Suppose $\hat{\theta}$ is an estimator of θ
  - $\hat{\theta} = g(X_1, \ldots, X_n)$
  - How does one conclude whether a given estimator is good?
- One metric for this is to evaluate $\left(\hat{\theta} - \theta\right)^2$
  - This is a random variable
  - Mean squared error: $\mathrm{MSE}(\hat{\theta}) = \mathbb{E}\left[\left(\hat{\theta} - \theta\right)^2\right]$
- We desire estimators with low mean-squared error

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Bias and Variance

- Algebraic identity: $\mathrm{MSE}(\hat{\theta}) = \mathbb{E}\left[\left(\hat{\theta} - \mathbb{E}[\hat{\theta}]\right)^2\right] + \left(\mathbb{E}[\hat{\theta}] - \theta\right)^2$

variance

Square of bias

- Suppose θ is the population mean we wish to estimate.
  - An estimator that outputs a constant c has a bias $c - \theta$ (zero variance; unbiased only if θ equals c)
  - An estimator that outputs X_i (for some i) will have no bias, but it will inherit the variance of X_i and its variance does not decrease as the sample size increases

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Slope estimator B $= \frac{\sum_{i=1}^{n}(x_i-\bar{x})(Y_i-\bar{Y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2}$

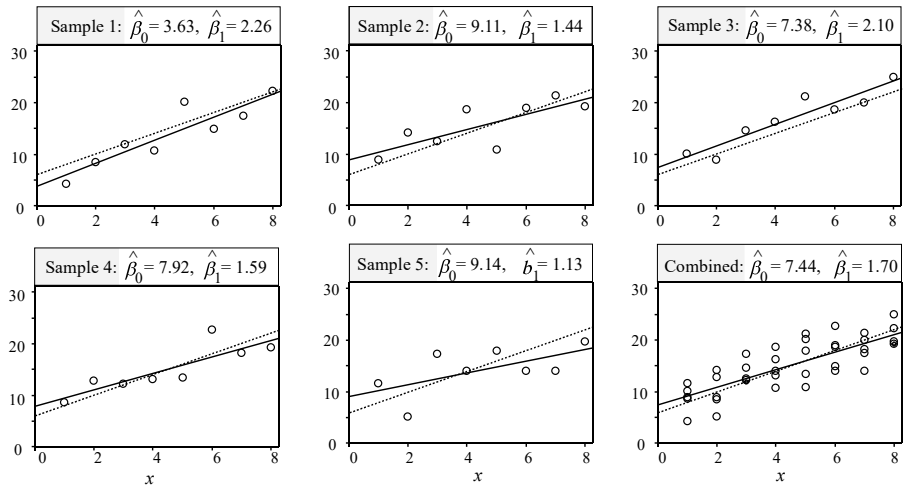
Intercept estimator A $= \bar{Y} - B\bar{x}$

Linear Regression Estimator

Notation from book: The A and B are estimators of parameters α and β in $Y = \alpha + \beta x + \epsilon$. They are random variables

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Example: Sampling variation

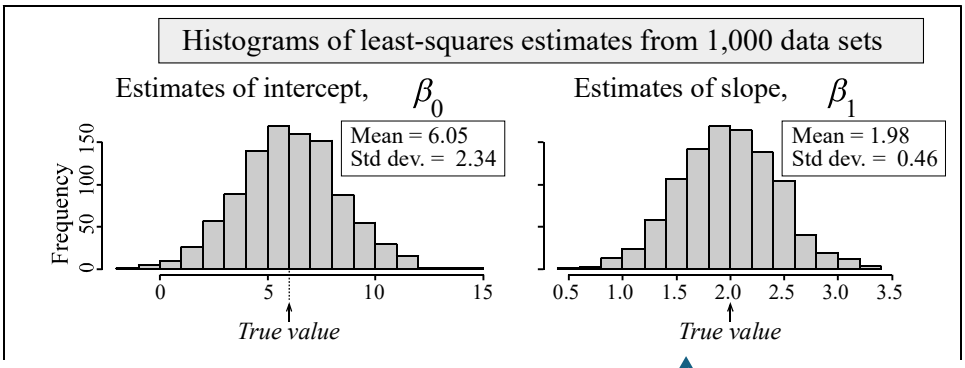


Notation:
$\hat{\beta}_0 = A$ and $\hat{\beta}_1 = B$
(from Ross book)

Data generated from $Y = 6 + 2x + \epsilon, \epsilon \sim \mathcal{N}(0,9)$

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Example: Distribution of estimator



Notation:
$\hat{\beta}_0 = A$ and $\hat{\beta}_1 = B$
(from Ross book)

intercept

slope

Data generated from $Y = 6 + 2x + \epsilon, \epsilon \sim \mathcal{N}(0,9)$

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Linear regression Parameters

- estimate $= \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2}, Y_i = \alpha + \beta x_i + \varepsilon_i, \varepsilon_i \sim N(0,\sigma^2)$
- $B = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(Y_i-\bar{Y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2} \Longrightarrow B = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(Y_i-\bar{Y})}{\sum_{j=1}^{n}(x_j-\bar{x})^2}$ ← Note the denominator
  - But $\sum_{i=1}^{n}(x_i-\bar{x})(Y_i-\bar{Y}) = \sum_{i=1}^{n}(x_i-\bar{x})Y_i - \bar{Y}\sum_{i=1}^{n}(x_i-\bar{x})$
- $B = \frac{\sum_{i=1}^{n}(x_i-\bar{x})Y_i}{\sum_{j=1}^{n}(x_j-\bar{x})^2} = \sum_{i=1}^{n}\frac{(x_i-\bar{x})}{\sum_{j=1}^{n}x_j-\bar{x})^2}Y_i$ (a linear combination of Y_i)
- $B \sim \mathcal{N}\left(\beta, \frac{\sigma^2}{\sum_i x_i^2 - n\bar{x}^2}\right)$

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Linear regression Parameters

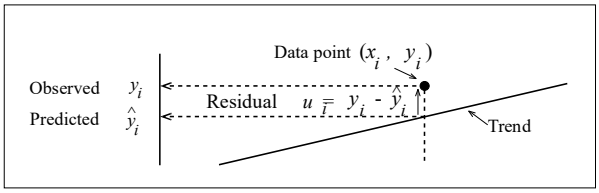
- $B \sim \mathcal{N}\left(\beta, \frac{\sigma^2}{\sum_i x_i^2 - n\bar{x}^2}\right)$
  - Unbiased estimator: $\mathbb{E}[B] = \beta$
- $A = \bar{Y} - B\bar{x}, A \sim \mathcal{N}(\alpha, \mathrm{Var}(A))$
  - Unbiased estimator: $\mathbb{E}[A] = \alpha$ and $\mathrm{Var}(A) = \frac{\sigma^2 \sum_i x_i^2}{n(\sum_i x_i^2 - n\bar{x}^2)}$

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Linear regression Parameters

- Define $\mathrm{SSR} = \sum_{i=1}^{n}(Y_i - A - B\mathrm{x}_i)^2$
  - Then $\frac{\mathrm{SSR}}{\sigma^2} \sim \chi^2_{n-2}, \Longrightarrow \mathbb{E}[\frac{SS_R}{\sigma^2}] = n - 2$
  - Unbiased: $\mathbb{E}\left[\frac{\mathrm{SSR}}{n-2}\right] = \sigma^2$

Note the denominator



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Least-squares fitting as MLE

- Consider the linear model
  - $Y_i = \beta x_i + \alpha + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$
  - $p(y_i \mid x_i, \beta, \alpha) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(y_i - \beta x_i - \alpha)^2}{2\sigma^2}\right]$
  - $\ell(\beta, \alpha, \sigma) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta x_i - \alpha)^2 - n\log\left(\sigma\sqrt{2\pi}\right)$
  - Minimizing the negative log likelihood results in
    - Same expression for A and B as in the ordinary least square (linear algebra)
    - Different expression for σ^2: $\boxed{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \mathrm{B}x_i - \mathrm{A})^2}$

Note the denominator

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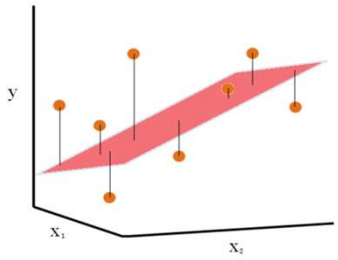
Review

- OLS and Gaussian MLE give the same coefficient estimators A and B.
- σ^2 is estimated from the residuals in both cases. They differ only in the divisor: n (MLE) or n−2 (unbiased).
  - The n−2 is the degrees of freedom left after fitting 2 parameters, the direct generalization of n−1.
- Normality buys exact sampling distributions and inference, not the point estimates.
- For large n the divisor difference is small. Use S^2=SSR/(n−2); software reports its observed value s^2.

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Agenda

- Earlier
  - Distribution of the estimators
  - Unknown variance
- Today
  - Multiple Linear regression



2

Least-squares optimization

- Simple linear model: $Y = \alpha + \beta x + \epsilon$, ϵ a random variable $\mathbb{E}[\epsilon] = 0$
  - Training data: $\{(x_i, y_i): i = 1 \ldots n\}$
  - α and β unknown, estimated using estimators A and B
- Want to minimize $\sum_{i=1}^{n}(Y_i - Bx_i - A)^2$ which is done either via calculus or linear algebra
- Result in the normal equations (estimator form)
- Observed-data form of solution
- One can show the estimators are *unbiased*

$$\sum_{i=1}^{n} x_i Y_i = B\sum_{i=1}^{n} x_i^2 + A\sum_{i=1}^{n} x_i,$$

$$\sum_{i=1}^{n} Y_i = B\sum_{i=1}^{n} x_i + nA$$

$$b_1 := b = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{(\sum_{i=1}^{n} x_i y_i) - n\bar{x}\bar{y}}{(\sum_{i=1}^{n} x_i^2) - n\bar{x}^2} \left(= r\frac{s_Y}{s_X}\right) \qquad b_0 := a = \bar{y} - b_1\bar{x}$$

3

Least-squares via normality assumption

- To find how good the estimator is we make additional assumptions on the error
  - $Y_i = \beta x_i + \alpha + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$
  - Likelihood: $p(y_i \mid x_i, \beta, \alpha) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(y_i - \beta\, x_i - \alpha)^2}{2\sigma^2}\right]$
  - Log likelihood $\ell(\beta, \alpha, \sigma) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta x_i - \alpha)^2 - n\log\left(\sigma\sqrt{2\pi}\right)$
- We reach the same equation as before for estimating α and β
- With $S_{xx} = \sum_i \mathrm{x}_i^2 - n\bar{\mathrm{x}}^2$, the distributions are obtained:
  - $B \sim \mathcal{N}(\beta, \frac{\sigma^2}{S_{xx}})$ and $A \sim \mathcal{N}(\alpha, \frac{\sigma^2 \sum_i \mathrm{x}_i^2}{nS_{xx}})$

4

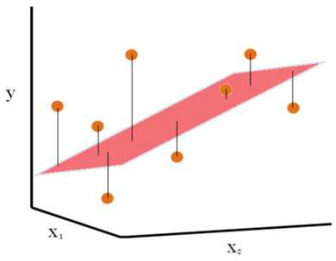
Variance

- Define $\mathrm{SSR} = \sum_{i=1}^{n}(Y_i - A - B\mathrm{x}_i)^2$
- With $S^2 = \frac{\mathrm{SSR}}{n-2}$, we have $\mathbb{E}[S^2] = \sigma^2$ (unbiased) (requires uncorrelated ϵ_i)
- With additional assumption of normality
  - biased MLE estimator for σ^2: $\boxed{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \mathrm{B}x_i - \mathrm{A})^2}$
  - Distribution $\frac{\mathrm{SSR}}{\sigma^2} \sim \chi^2_{n-2}$

5

Multiple Linear Regression

- $Y = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \varepsilon$
  - Training data: $\{(x_{i1}, x_{i2}, \dots x_{ik}, y_i): i = 1 \dots n\}$
  - $B = \arg\min_\beta \sum_{i=1}^{n} \left(Y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2$
  - $B=(B_0,\dots,B_k)^T$ estimates $\beta=(\beta_0,\dots,\beta_k)^T$



7

Multiple Linear Regression via normality

- $Y = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \varepsilon$
  - Training data: $\{(x_{i1}, x_{i2}, \dots x_{ik}, y_i): i = 1 \dots n\}$
  - $Y \mid x_1, \dots, x_k \sim \mathcal{N}(\mu_x, \sigma^2)$
    - $\mu_x = \beta_0 + \sum_{j=1}^{k} \beta_j x_j$
- $L(\beta_0, \dots, \beta_k, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{\left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2}{2\sigma^2}\right]$
- $\ell = -\frac{1}{2\sigma^2} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2 - \frac{n}{2} \log(2\pi\sigma^2)$
- Applying the maximizing rule to random responses gives the estimator $B = \arg\min_\beta \sum_{i=1}^{n} \left(Y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2$

8

Normal Equations

$$\ell(\beta_0, \ldots, \beta_k, \sigma^2) = -\frac{1}{2\sigma^2} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij} \right)^2 - \frac{n}{2} \log(2\pi\sigma^2)$$

$$\frac{\partial \ell}{\partial \beta_0} = \frac{1}{\sigma^2} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij} \right) = 0$$

$$\frac{\partial \ell}{\partial \beta_1} = \frac{1}{\sigma^2} \sum_{i=1}^{n} x_{i1} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij} \right) = 0$$

$$\frac{\partial \ell}{\partial \beta_r} = \frac{1}{\sigma^2} \sum_{i=1}^{n} x_{ir} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij} \right) = 0, \quad r = 1, \ldots, k$$

Replacing the observed responses y_i by the random responses Y_i makes the minimizing coefficients the estimators B_0 … B_k (next slide)

9

Normal Equations

$$\sum_{i=1}^{n} Y_i = B \sum_{i=1}^{n} x_i + nA$$

$$\sum_{i=1}^{n} x_i Y_i = B \sum_{i=1}^{n} x_i^2 + A \sum_{i=1}^{n} x_i$$

Earlier simple linear regression

$$\sum_{i=1}^{n} Y_i = nB_0 + B_1 \sum_{i=1}^{n} x_{i1} + B_2 \sum_{i=1}^{n} x_{i2} + \cdots + B_k \sum_{i=1}^{n} x_{ik}$$

$$\sum_{i=1}^{n} x_{i1} Y_i = B_0 \sum_{i=1}^{n} x_{i1} + B_1 \sum_{i=1}^{n} x_{i1}^2 + B_2 \sum_{i=1}^{n} x_{i1} x_{i2} + \cdots + B_k \sum_{i=1}^{n} x_{i1} x_{ik}$$

$$\sum_{i=1}^{n} x_{ik} Y_i = B_0 \sum_{i=1}^{n} x_{ik} + B_1 \sum_{i=1}^{n} x_{ik} x_{i1} + B_2 \sum_{i=1}^{n} x_{ik} x_{i2} + \cdots + B_k \sum_{i=1}^{n} x_{ik}^2$$

11

Multiple Linear Regression

- $Y = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \varepsilon$
  - Training data: $\{(x_{i1}, x_{i2}, \ldots x_{ik}, y_i): i = 1 \ldots n\}$
- Rewrite all the n equations in matrix form

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \qquad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix} \qquad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} \qquad \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

$$Y \in \mathbb{R}^{n\times 1}, \quad X \in \mathbb{R}^{n\times p}, \quad \beta \in \mathbb{R}^{p\times 1}, \quad \epsilon \in \mathbb{R}^{n\times 1} \qquad p = k + 1$$

$$Y = X\beta + \epsilon$$

This is now a matrix notation

12

Assemble Matrices

$$X^T = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_{11} & x_{21} & \cdots & x_{n1} \\ x_{12} & x_{22} & \cdots & x_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1k} & x_{2k} & \cdots & x_{nk} \end{bmatrix} \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix} \quad Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad X^TY = \begin{bmatrix} \sum_i Y_i \\ \sum_i x_{i1}Y_i \\ \vdots \\ \sum_i x_{ik}Y_i \end{bmatrix}$$

$$X^TX = \begin{bmatrix} n & \sum_i x_{i1} & \sum_i x_{i2} & \cdots & \sum_i x_{ik} \\ \sum_i x_{i1} & \sum_i x_{i1}^2 & \sum_i x_{i1}x_{i2} & \cdots & \sum_i x_{i1}x_{ik} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum_i x_{ik} & \sum_i x_{ik}x_{i1} & \sum_i x_{ik}x_{i2} & \cdots & \sum_i x_{ik}^2 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} B_0 \\ B_1 \\ \vdots \\ B_k \end{bmatrix}$$

13

Normal Equations

$$\sum_{i=1}^{n} Y_i = nB_0 + B_1 \sum_{i=1}^{n} x_{i1} + B_2 \sum_{i=1}^{n} x_{i2} + \cdots + B_k \sum_{i=1}^{n} x_{ik}$$

$$\sum_{i=1}^{n} x_{i1} Y_i = B_0 \sum_{i=1}^{n} x_{i1} + B_1 \sum_{i=1}^{n} x_{i1}^2 + B_2 \sum_{i=1}^{n} x_{i1} x_{i2} + \cdots + B_k \sum_{i=1}^{n} x_{i1} x_{ik}$$

$$\sum_{i=1}^{n} x_{ir} Y_i = B_0 \sum_{i=1}^{n} x_{ir} + \sum_{j=1}^{k} B_j \sum_{i=1}^{n} x_{ir} x_{ij}, \quad r = 1, \ldots, k$$

$$\sum_{i=1}^{n} x_{ik} Y_i = B_0 \sum_{i=1}^{n} x_{ik} + B_1 \sum_{i=1}^{n} x_{ik} x_{i1} + B_2 \sum_{i=1}^{n} x_{ik} x_{i2} + \cdots + B_k \sum_{i=1}^{n} x_{ik}^2$$

$$X^T Y = X^T X B$$

$$B = (X^T X)^{-1} X^T Y$$

Assumes invertibility

14

Example

- Diameter of a tree is measured at a standardized point and is a function of the age, elevation, rainfall and specific gravity
- We can, for instance, ignore the specific gravity and compute the estimated linear regression model to be $y = 11.54873 + 0.05728x_1 + 0.08712x_2 + 7.33231x_3$



	Age (years)	Elevation (1,000 ft)	Rainfall (inches)	Specific Gravity	
1	44	1.3	250	.63	18.1
2	33	2.2	115	.59	19.6
3	33	2.2	75	.56	16.6
4	32	2.6	85	.55	16.4
5	34	2.0	100	.54	16.9
6	31	1.8	75	.59	17.0
7	33	2.2	85	.56	20.0
8	30	3.6	75	.46	16.6
9	34	1.6	225	.63	16.2
10	34	1.5	250	.60	18.5
11	33	2.2	255	.63	18.7
12	36	1.7	175	.58	19.4
13	33	2.2	75	.55	17.6
14	34	1.3	85	.57	18.3
15	37	2.6	90	.62	18.8

15

Estimators are unbiased

$$
\begin{aligned}
E[\mathbf{B}] &= E[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}] \\
&= E[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{X}\boldsymbol{\beta}+\mathbf{e})] \quad \text{since } \mathbf{Y}=\mathbf{X}\boldsymbol{\beta}+\mathbf{e} \\
&= E[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta}+(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{e}] \\
&= E[\boldsymbol{\beta}+(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{e}] \\
&= \boldsymbol{\beta}+(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top E[\mathbf{e}] \\
&= \boldsymbol{\beta}
\end{aligned}
$$

16

Variance

- Define SSR as before as the sum of the squares of the residuals
  - $\Sigma_i(Y_i - B_o - \Sigma_j B_j x_{ij})^2$
- With $S^2 = \frac{\text{SSR}}{n-\text{k}-1}$, we have $\mathbb{E}[S^2] = \sigma^2$ (unbiased)

17

Multivariate Normal Distributions

- The least square estimators $B_0, B_1, \ldots B_k$ which are the elements of the matrix B are linear combinations of the independent normal random variables Y_i
- The set of independent random variables form an interesting object
  - The multivariate normal distribution
- We say the estimator vector B has a multivariate normal distribution

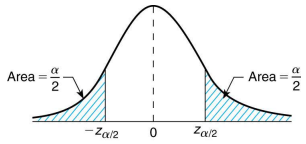
$$B = (X^T X)^{-1} X^T Y$$

18

Agenda

- Earlier
  - Confidence intervals

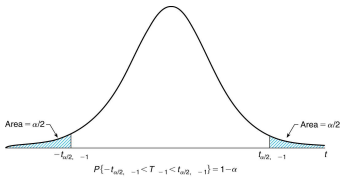
$$\Pr\left(-z_{0.025} \leq \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \leq z_{0.025}\right) = 0.95$$



1

Review: 95% confidence interval

- When we estimate a population parameter θ using an estimate $\hat{\theta}$, the defining property of a 95% confidence interval is
  - "a type of interval that contains the true value of θ for 95% of samples"
- In our examples: a confidence interval is the $\hat{\theta} \pm t$ standard errors
- This works exactly if the distribution is normal, and
  - "quite well" when the distribution is unknown
  - $(\bar{X}-\mu)/(S/\sqrt{n}) \sim t_{n-1}$
  - $\Pr\left(-t_{0.025,n-1} \leq \frac{\bar{X}-\mu}{S/\sqrt{n}} \leq t_{0.025,n-1}\right) = 0.95,$



2

Review

TABLE 7.1 *100(1 −α) Percent Confidence Intervals*

$$X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$$

$$\overline{X} = \sum_{i=1}^{n} X_i/n, \qquad S = \sqrt{\sum_{i=1}^{n} (X_i - \overline{X})^2/(n-1)}$$

Assumption	Parameter	Confidence Interval	Lower Interval	Upper Interval
σ^2 known	μ	$\overline{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	$\left(-\infty, \overline{X} + z_{\alpha}\frac{\sigma}{\sqrt{n}}\right)$	$\left(\overline{X} + z_{\alpha}\frac{\sigma}{\sqrt{n}}, \infty\right)$
σ^2 unknown	μ	$\overline{X} \pm t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}$	$\left(-\infty, \overline{X} + t_{\alpha,n-1}\frac{S}{\sqrt{n}}\right)$	$\left(\overline{X} - t_{\alpha,n-1}\frac{S}{\sqrt{n}}, \infty\right)$

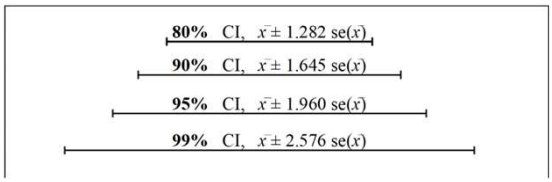
We use the empirical standard deviation in practice

Lower bound: at least as large

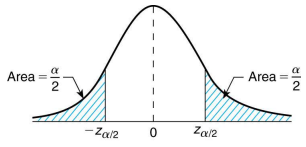
3

Agenda

- Earlier
  - Confidence intervals
- Today
  - Comparing two populations or treatments



$$\Pr\left(-z_{0.025} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{0.025}\right) = 0.95$$



4

Course details from IITB for CS215

- The role of statistics.
- Graphical and numerical methods for describing and summarising data.
- Probability.
- Population distributions.
- Sampling variability and sampling distributions.
- Estimation using a single sample.
- Hypothesis testing a single sample.
- *Comparing two populations or treatments.*
- Simple linear regression and correlation.
- Case studies.

Example: Comparing two populations

- Rheumatoid arthritis (RA) is a debilitating condition
- We measuring sulfhydryl concentration (mmol) in blood samples
- Statistical question:
  - What is the difference between concentration in those without and with RA?
  - Let's compute confidence intervals

	Without RA	With RA
	1.84	1.765
	1.92	3.015
	1.94	2.575
	1.92	2.225
	1.85	2.225
	1.91	2.715
	2.07	
Sample size	7	6
Sample mean	1.92143	2.42000
Sample standard deviation	0.07559	0.44049

5

Example: Comparing two populations

- Statistical question: what is the difference between those without RA and those with rheumatoid arthritis?
  - n = 7, df = 6, t(0.025, 6) = 2.447 SE = 0.07559/√7 = 0.02857, margin = 2.447 × 0.02857 = 0.0699,CI = 1.92143 ± 0.0699 = (1.852, 1.991)
  - n = 6, df = 5, t(0.025, 5) = 2.571 SE = 0.44049/√6 = 0.17983, margin = 2.571 × 0.17983 = 0.4623 CI = 2.42 ± 0.4623 = (1.958, 2.882)
- The intervals overlap.
  - This might indicate that, for the without RA condition, the true mean is more than 1.958 (lower end of the with-RA condition)
  - This is not the way we compare two populations

6

How different are the two population?

- Confidence interval for a difference between population means
  - $(\mu_1 - \mu_2)$: $(\bar{x}_1 - \bar{x}_2) \pm t\ \mathrm{se}(\bar{x}_1 - \bar{x}_2)$ ← Two means
- When two samples are independent the variance adds up, so

  $\mathrm{se}(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
- Conservative degree of freedom approximation for the t-distribution:
  - $\min(n_1 - 1, n_2 - 1)$ ← Employed in practice
- In the RA blood sample example:
  - Difference between means = 2.42 – 1.92143 = 0.49857, se = sqrt(0.07559²/7 + 0.44049²/6) = 0.1821, t(0.025, 5) = 2.571, margin=0.4681, CI = (0.030, 0.967) does not include zero, the with-RA always exceeds the without-RA

7

Confidence intervals for difference

- Previous discussions assume empirical approximations
- If population is normal and means are estimated,
  - Assume: variances are known, then CI uses the known variances, they are added, and we work with exact normal distribution (see next slide)
  - Assume variances are unknown, and we use the empirical values
    - If they are different, then CI is complicated (o-o-s).
    - If they are same, then CI uses the pooled estimator and df = (n+m-2)
  - Note: Many cases variances are unknown and different!

8

Review

TABLE 7.2 *100(1 − σ) Percent Confidence Intervals for $\mu_1 - \mu_2$*

$$X_1, \ldots, X_n \sim \mathcal{N}(\mu_1, \sigma_1^2)$$
$$Y_1, \ldots, Y_m \sim \mathcal{N}(\mu_2, \sigma_2^2)$$
$$\overline{X} = \sum_{i=1}^{n} X_i/n, \qquad S_1^2 = \sum_{i=1}^{n} (X_i - \overline{X})^2/(n-1)$$
$$\overline{Y} = \sum_{i=1}^{m} Y_i/n, \qquad S_2^2 = \sum_{i=1}^{m} (Y_i - \overline{Y})^2/(m-1)$$

Assumption	Confidence Interval
σ_1, σ_2 known	$\overline{X} - \overline{Y} \pm z_{\alpha/2}\sqrt{\sigma_1^2/n + \sigma_2^2/m}$
σ_1, σ_2 unknown but equal	$\overline{X} - \overline{Y} \pm t_{\alpha/2, n+m-2}\sqrt{\left(\frac{1}{n} + \frac{1}{m}\right)\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}$

Pooled estimator

Assumption	Lower Confidence Interval
σ_1, σ_2 known	$(-\infty, \overline{X} - \overline{Y} + z_{\alpha}\sqrt{\sigma_1^2/n + \sigma_2^2/m})$
σ_1, σ_2 unknown but equal	$\left(-\infty, \overline{X} - \overline{Y} + t_{\alpha, n+m-2}\sqrt{\left(\frac{1}{n} + \frac{1}{m}\right)\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}\right)$

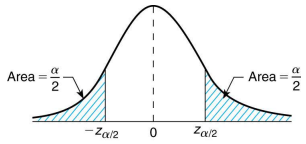
Note: Upper confidence intervals for $\mu_1 - \mu_2$ are obtained from lower confidence intervals for $\mu_2 - \mu_1$.

9

Agenda

- Earlier
  - Confidence intervals for proportions
  - Comparing means for two populations or treatments
- Today
  - Confidence intervals for proportions from two different populations

$$\Pr\left(-z_\alpha \leq \frac{\hat{P}-p}{\sqrt{\frac{p(1-p)}{n}}} \leq z_\alpha\right)$$



1

Review

2

Example: Comparing two populations

- We measuring sulfhydryl concentration (mmol) in blood samples of those without and with rheumatoid arthritis (RA)
- Statistical question:
  - What is the difference in the concentration for those without and with RA?
  - We look for the confidence interval in the difference in means using the provided data

	Without RA	With RA
	1.84	1.765
	1.92	3.015
	1.94	2.575
	1.92	2.225
	1.85	2.225
	1.91	2.715
	2.07	
Sample size	7	6
Sample mean	1.92143	2.42000
Sample standard deviation	0.07559	0.44049

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- Hypothesis testing a single sample.
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- Case studies.

3

Review: Confidence interval for difference

TABLE 7.2 *100(1 − σ) Percent Confidence Intervals for* $\mu_1 - \mu_2$

$$X_1, \ldots, X_n \sim \mathcal{N}(\mu_1, \sigma_1^2)$$
$$Y_1, \ldots, Y_m \sim \mathcal{N}(\mu_2, \sigma_2^2)$$
$$\overline{X} = \sum_{i=1}^{n} X_i/n, \qquad S_1^2 = \sum_{i=1}^{n} (X_i - \overline{X})^2/(n-1)$$
$$\overline{Y} = \sum_{i=1}^{m} Y_i/n, \qquad S_2^2 = \sum_{i=1}^{m} (Y_i - \overline{Y})^2/(m-1)$$

Assumption	Confidence Interval
σ_1, σ_2 known	$\overline{X} - \overline{Y} \pm z_{\alpha/2}\sqrt{\sigma_1^2/n + \sigma_2^2/m}$
σ_1, σ_2 unknown but equal	$\overline{X} - \overline{Y} \pm t_{\alpha/2,\, n+m-2}\sqrt{\left(\frac{1}{n} + \frac{1}{m}\right)\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}$

Pooled estimator

If we do not know the variances, and whether the variances are equal, then, as an empirical measure, we use a conservative estimate with df=min(n-1, m-1)

4

Review: Proportions

CI is not a protection against fabricated or sloppy data. We assume random sampling with independent observations, and an unbiased instrument to measure the data

- When we estimate a population parameter θ using an estimate $\hat{\theta}$, the defining property of a 95% confidence interval is
  - "a type of interval that contains the true value of θ for 95% of samples"
- In our examples: a confidence interval is the $\hat{\theta} \pm$ z standard errors applied to *proportions*
- This works well if the distribution is normal, and
  - "quite well" when the distribution is unknown provided n is "sufficiently" large
  - Example: np(1-p)>10

$$\Pr\left(-z_{0.025} \leq \frac{\hat{p}-p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \leq z_{0.025}\right) \approx 0.95$$

5

Example: Comparing two populations

- 68% (n=870) non-smokers get "good" grades compared to 41% (n=130) smokers.
- What can we conclude?
- Contrast a single population where we ask
  - Does the fraction 0.68 (for non-smokers) represent the true proportion in the population

Get good grades

percent

Characteristics by Smoking and Drinking Status

(Table entry is % of group saying yes)	Smoker	Nonsmoker
Get mostly A's or B's?	41	68
Read 1 or more hours/day?	54	72
Get drunk at least once a month?	63	10
Have smoked Marijuana?	79	14
Likely to try illegal drug in future?	42	14
n	130	870

Smoked cigarettes in last 30 days.

Consumed more than a few sips of alcohol in the last 30 days.

Source: "Back to School Teen Survey"

6

Review: Single population

$$\Pr\left(-z_\alpha \le \frac{\hat{P}-p}{\sqrt{\frac{p(1-p)}{n}}} \le z_\alpha\right)$$

- Does the fraction 0.68 (for non-smokers) represent the true proportion in the population?
  1. Is the normal approximation valid? 870*0.68*(1-0.68) > 10, so yes. Empirical rule
  2. $\Pr\left(-z_{0.025} \le \frac{\hat{p}-p}{se(\hat{p})} \le z_{0.025}\right) = 0.95$ and $z_{0.025} = 1.96$,
  3. $se(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.01582$
  4. Therefore $CI = \hat{p} \pm 1.96\,\widehat{se}(\hat{p})$
     $= 0.68 \pm 1.96(0.01582)$
     $= 0.68 \pm 0.03100$
  5. It is not unreasonable to expect 65% of non-smokers getting good grades in the population

Non-causality: Non-smoking is not a shortcut to getting good grades

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Review: How big should my sample be?

- The margin is $z \times \sqrt{(}p(1-p)/n$
- If m is the margin of error, then
  - want $z \times \sqrt{(}p(1-p)/n \le m$
  - Implies if we guess a proportion p^* want $n \approx \frac{z}{m}^2 p^*(1-p^*)$
- Examples
  - Margin is 10%, p=0.3, n=81
  - Margin is 5%, p=0.3, n=323

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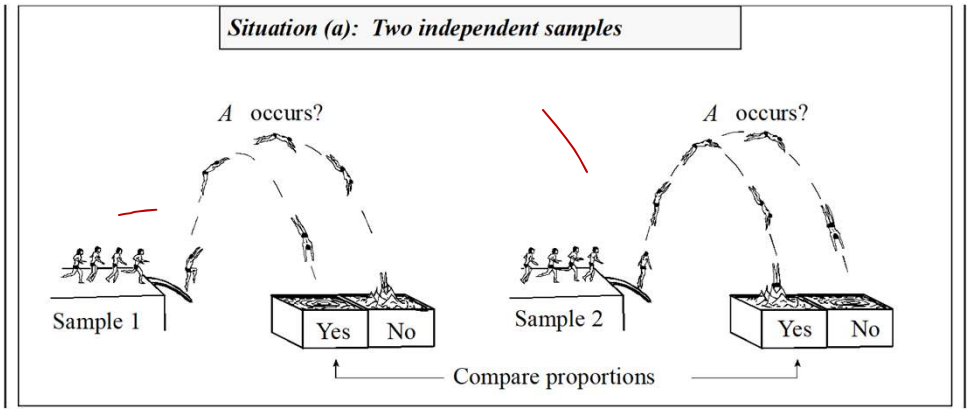
Example: Comparing two populations

- Good grades: 68% (n=870) non-smokers, 41% (n=130) smokers
- What can we conclude about the difference between the two populations?
  - Is the normal approximation tenable?
    - 130*0.41*(1-0.41) > 10? 870*0.68*(1-0.68) > 10, so yes.
  - Point estimate of the difference = 0.68-0.41, standard error = 0.045944
  - CI = $0.27 \pm 1.96 * 0.045944$= [0.18, 0.36]
  - Not unreasonable that non-smokers outperform smokers, the difference is 18 percentage points (with 95% confidence)

$$se(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

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Proportions from independent samples



$$se(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Variance adds up

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Example #2

- 18% difference in the level of support for Clinton. What's the CI?
- Level of support for Clinton in NJ and NY: may be considered independent
- Level of support for Clinton vs Dole in New Jersey: same population is being asked to make a choice
  - What's the difference?

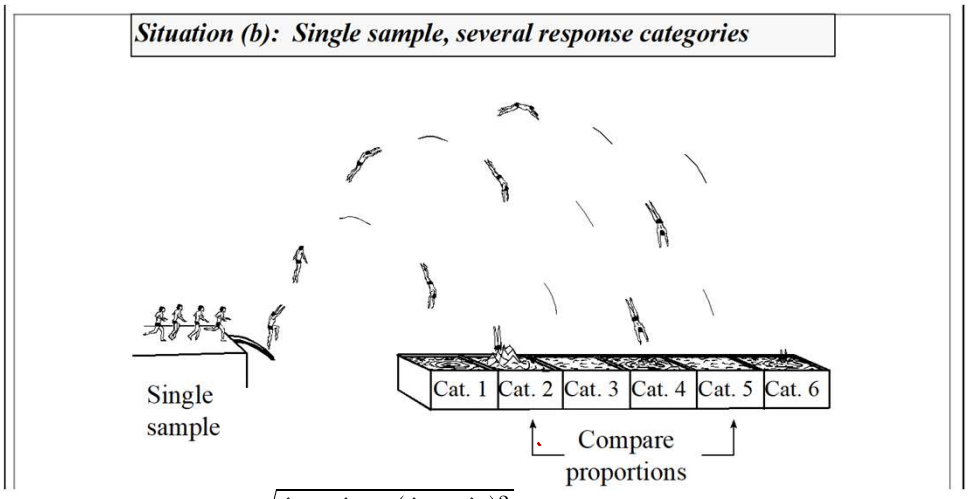
percent

State	n	Pre-election Polls				Election Results		
		Clinton	Dole	Perot	Other/Undecided	Clinton	Dole	Perot
New Jersey	1,000	51	33	8	8	53	36	9
New York	1,000	59	25	7	9	59	31	8
Connecticutt	1,000	51	29	11	9	52	35	10

Source: Quinnipiac College Polling Institute press release, 8 Nov. 1996.

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Proportions



$$\text{se}(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1 + \hat{p}_2 - (\hat{p}_1 - \hat{p}_2)^2}{n}}$$

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Example

- Level of support: for Clinton: 51 and for Dole: 33, difference is 18%.
- What is the confidence interval?
  - Answer: margin = 0.056, CI = [0.12, 0.24]

State	n	Pre-election Polls				Election Results		
		Clinton	Dole	Perot	Other/Undecided	Clinton	Dole	Perot
New Jersey	1,000	51	33	8	8	53	36	9
New York	1,000	59	25	7	9	59	31	8
Connecticutt	1,000	51	29	11	9	52	35	10

Source: Quinnipiac College Polling Institute press release, 8 Nov. 1996 .

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Example #3

- Consider smokers who admit to getting drunk at least once a month (63%) vs smoked marijuana (79%)
  - Are smokers more likely to try marijuana compared to drinking?

Characteristics by Smoking and Drinking Status

(Table entry is % of group saying yes)	Smoker	Nonsmoker
Get mostly A's or B's?	41	68
Read 1 or more hours/day?	54	72
Get drunk at least once a month?	63	10
Have smoked Marijuana?	79	14
Likely to try illegal drug in future?	42	14
n	130	870

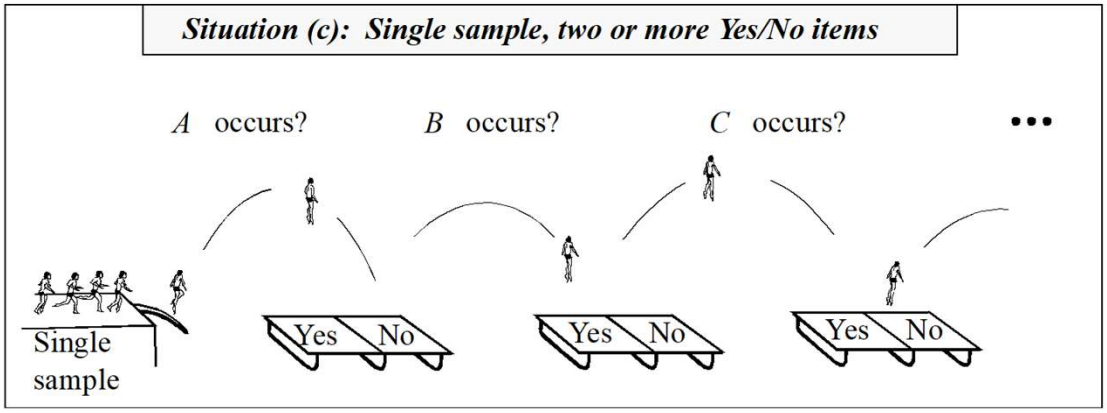
Smoked cigarettes in last 30 days.

Consumed more than a few sips of alcohol in the last 30 days.

Source: "Back to School Teen Survey"

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Proportions from independent samples



$$se(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\min(\hat{p}_1 + \hat{p}_2, \hat{q}_1 + \hat{q}_2) - (\hat{p}_1 - \hat{p}_2)^2}{n}} \qquad \hat{q}_1 = 1 - p_1, \hat{q}_2 = 1 - p_2$$

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Example #3

- Consider smokers who admit to getting drunk at least once a month (63%) vs smoked marijuana (79%)
  - Are smokers more likely to try marijuana compared to drinking?
- Point estimate = 0.79 – 0.63
- n = 130 is large enough
- se = 0.065, [0.03, 0.29]

Characteristics by Smoking and Drinking Status

(Table entry is % of group saying yes)		Smoker	Nonsmoker
Get mostly A's or B's?		41	68
Read 1 or more hours/day?		54	72
Get drunk at least once a month?		63	10
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Likely to try illegal drug in future?		42	14
	n	130	870

Smoked cigarettes in last 30 days.

Consumed more than a few sips of alcohol in the last 30 days.

Source: "Back to School Teen Survey"

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Review

1. First verify if the normal approximation can be made
2. Check which case applies
   - Proportions from two independent samples
   - One sample of size n, several response categories
   - One sample of size n, many yes/no items
   - Calculate the standard error based on the formula
   - Use the normal approximation with the appropriate confidence level

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